

Nephrotic Syndrome: Primary and Secondary

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CHAPTER OUTLINE

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KEY POINTS

- Nephrotic syndrome is a clinical condition arising from podocyte injury, typically associated with glomerular diseases collectively termed podocytopathies. These disorders are characterized ultrastructurally by diffuse effacement of podocyte foot processes. Less commonly, nephrotic syndrome may result from other histologic patterns of injury, such as those seen in amyloidosis or diabetic kidney disease.
- Minimal change disease is the predominant cause of nephrotic syndrome in children (70%–90%) and accounts for 10% to 20% of adult cases. Most pediatric cases are managed without kidney biopsy and are termed idiopathic nephrotic syndrome. Some studies have identified minimal change disease/idiopathic nephrotic syndrome as an autoimmune disease, with nephrin as a target antigen and antinephrin antibodies detected in a substantial proportion of cases. It presents with abrupt nephrotic syndrome, responds rapidly to glucocorticoids, and has excellent kidney outcomes (<5% risk of end-stage kidney failure [ESKF]). Its relapsing nature, requiring repeated immunosuppression, underscores the need for therapies reducing glucocorticoid exposure and related complications.
- Focal segmental glomerulosclerosis accounts for 20% to 40% of adult nephrotic syndrome cases and arises from a highly variable spectrum of causes. These range from immune-mediated podocyte injury and genetic mutations affecting podocyte and nonpodocyte genes to secondary causes, including maladaptive response to glomerular hyperfiltration. The diverse underlying mechanisms leading to a common histologic pattern pose significant diagnostic challenges, complicating treatment strategies. Additionally, a considerable proportion of patients experience treatment resistance and progressive kidney function decline, making focal segmental glomerulosclerosis a major contributor to chronic kidney disease and ESKF globally.
- Membranous nephropathy (MN) is the leading histologic diagnosis in nondiabetic adult patients with nephrotic syndrome and one of the main causes of ESKF among glomerular diseases. It is an autoimmune disease driven by autoantibodies against podocyte antigens, with M-type phospholipase A2 receptor found in 75% of cases, and numerous newly identified antigens accounting for most phospholipase A2 receptor-negative cases, forming the basis for a proposed novel disease classification. Approximately 30% of MN patients achieve spontaneous remission, while 30% to 40% progress to ESKF within 10 years, and a similar proportion exhibits resistance to available therapies. Proper identification of these patient subsets and optimization of treatment strategies remain critical challenges in MN management.

Glomerular diseases comprise a heterogeneous group of conditions, representing the third leading cause of end-stage kidney failure (ESKF). These disorders affect various structural components of the renal glomeruli, particularly those that constitute the glomerular filtration barrier. This critical barrier includes capillary endothelial fenestrations, the glomerular basement membrane (GBM), and the interdigitating foot processes of podocytes, connected by slit diaphragms. Damage to these components manifests as distinct clinical syndromes depending on the affected structure.

Injury to podocytes results in severe symptomatic proteinuria, a hallmark of clinical phenotype called nephrotic syndrome (NS). Diseases primarily affecting podocytes are classified as podocytopathies. Conversely, injury to the GBM and endothelial cells, often secondary to inflammatory processes, leads to nephritic syndrome. This condition is marked by active urine sediment,

typified by hematuria, red blood cell casts, with or without accompanying and subnephrotic proteinuria. Disorders associated with nephritic syndrome are collectively termed glomerulonephritis. Nephrotic and nephritic syndromes represent two ends of the clinical spectrum of glomerular diseases (Fig. 30.1), though they frequently overlap in clinical presentations (eTable 30.1). Therefore renal biopsy remains the gold standard for diagnosis, providing essential information for accurate classification and guiding subsequent diagnostic and therapeutic decisions.

The hallmark histologic feature in podocytopathies is podocyte foot process effacement (FPE), typically observed as diffuse and global involvement, affecting nearly the entire surface area of the glomeruli, as seen on electron microscopy. Podocytopathies encompass a variety of injury patterns including minimal change disease (MCD), focal segmental glomerulosclerosis (FSGS), and membranous nephropathy (MN). Both MCD and FSGS are characterized

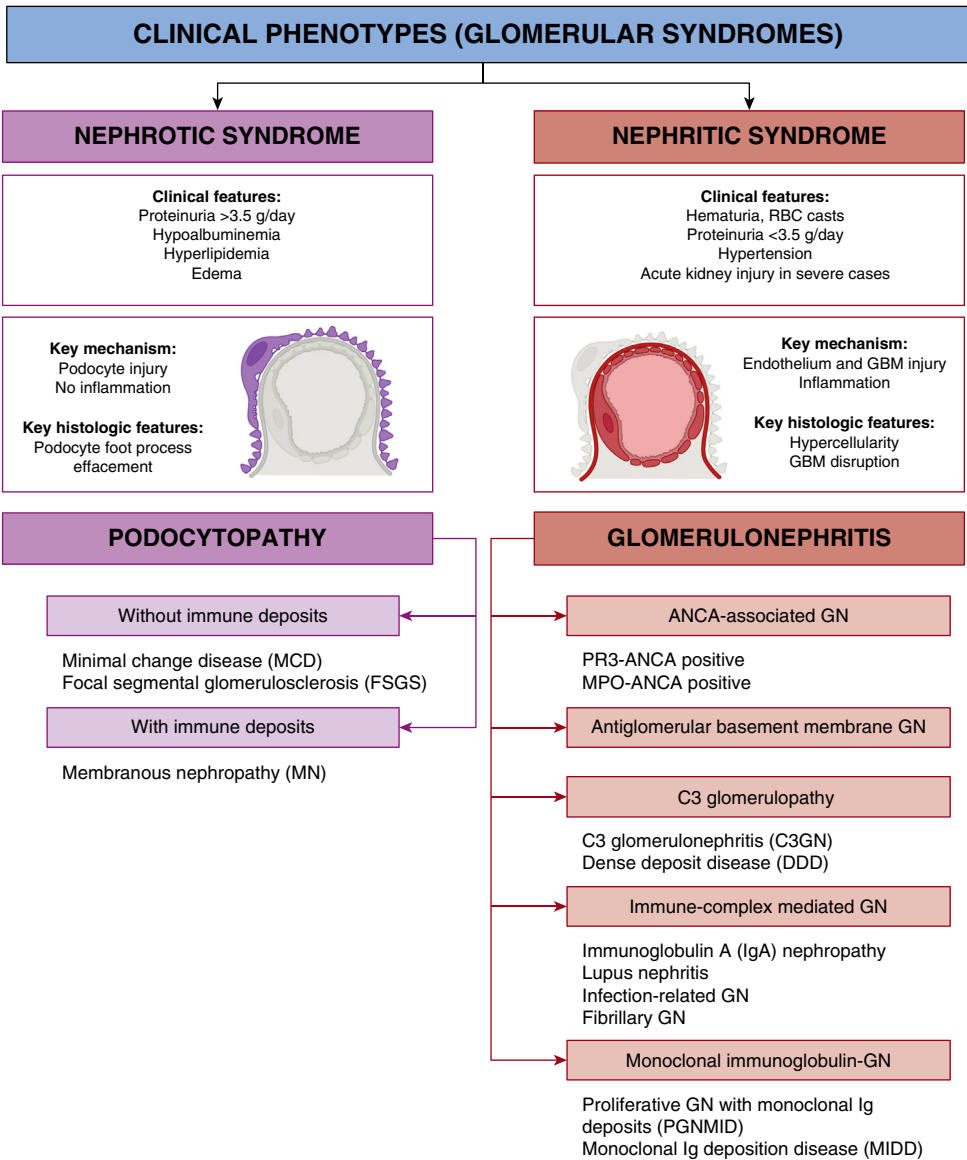


Fig. 30.1 Main clinical phenotypes in glomerular diseases (glomerular syndromes). ANCA, Antineutrophil cytoplasmic antibody; GBM, glomerular basement membrane; GN, glomerulonephritis.

eTable 30.1 Manifestations of Nephrotic and Nephritic Features by Glomerular Diseases

Disease	Nephrotic Features	Nephritic Features
Minimal change disease	++++	–
Membranous nephropathy	++++	+
Focal segmental glomerulosclerosis	+++	++
Fibrillary glomerulonephritis	+++	++
Mesangioproliferative ^a	++	++
Membranoproliferative glomerulonephritis ^b	++	+++
Proliferative glomerulonephritis ^a	++	+++
Acute diffuse proliferative glomerulonephritis ^c	+	++++
Crescentic glomerulonephritis ^d	+	++++

^aMesangioproliferative and proliferative glomerulonephritis (focal or diffuse) are structural manifestations of a number of glomerulonephritides, including IgA nephropathy and lupus nephritis.

^bBoth mesangiocapillary (type I) and dense deposit disease (type II).

^cOften a manifestation of acute poststreptococcal glomerulonephritis.

^dCan be immune-complex-mediated, anti-glomerular basement membrane antibody mediated, or associated with antineutrophil cytoplasmic autoantibodies.

Modified from Jennette JC, Mandal AK. The nephrotic syndrome. In: Mandal AK, Jennette JC, eds. *Diagnosis and Management of Renal Disease and Hypertension*. Durham, NC: Carolina

Box 30.1 Alterations of Plasma Protein and Procoagulant Factors in Nephrotic Syndrome**Plasma globulins**

- Decreased immunoglobulin G (IgG)
- Normal or increased IgA, IgM, or IgE
- Increased α_2 - and β -globulins
- Decreased α_1 -globulins

Metal- and hormone-binding proteins

- Loss of metal binding proteins: iron, copper, zinc
- Loss of erythropoietin, transferrin
- Loss of thyroid-binding globulin, vitamin D-binding protein, sex-hormone binding globulin, transcortin

Complement system

- Decreased factor B and I
- Decreased C1q, C2, C8, C9
- Increased C3
- Normal C1s, C4, and C1 inhibitor

Coagulation components

- Increased factors II, V, VII, VIII, X, XIII
- Decreased factors IX, XI, XII
- Increased plasma fibrinogen
- Increased intravascular fibrin formation
- Decreased fibrinolytic activity
- Decreased plasma plasminogen
- Increased tissue type plasminogen activator
- Increased α_2 -macroglobulin
- Decreased α_1 -antitrypsin
- Decreased antithrombin III
- Decreased protein S

Other procoagulant factors

- Increased blood viscosity
- Hemoconcentration
- Thrombocytosis
- Increased platelet aggregability

by the absence of immune-complex deposits, with MCD showing a normal glomerular appearance on light microscopy, while FSGS is marked by focal and segmental sclerotic lesions. In contrast, MN is distinguished by the presence of immune-complex deposits localized to the subepithelial zone of the GBM, accompanied by GBM thickening.

In contrast to podocytopathies, the hallmark of glomerulonephritis is the presence of inflammatory and proliferative lesions. These include endothelial injury and GBM disruption, as well as hypercellularity within the endocapillary, mesangial, and/or extracapillary compartments. These lesions arise through various pathogenetic mechanisms including anti-GBM disease, antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, immune-complex-mediated injury, C3 glomerulopathy, and monoclonal gammopathy, each with distinct etiologies and clinical presentations (as discussed in [Chapters 31–36](#)).

It is important to note that neither the clinical phenotype nor the histologic pattern of injury alone is sufficient to make a definitive diagnosis. A comprehensive diagnostic workup is essential to determine the underlying cause, as glomerular diseases may result from either an immune-mediated kidney-limited process (termed “primary”) or from a variety of secondary causes. These secondary causes can include autoimmune diseases, malignancy-associated conditions, infections, genetic mutations, and drug-induced nephrotoxicity, with kidney involvement representing one of several possible manifestations.

Advances have significantly improved our understanding of the target antigens involved in pathogenesis of glomerular diseases, particularly in podocytopathies. This expanding knowledge has enabled the development of more accurate and comprehensive diagnostic pathways, allowing for the identification of specific antigens in an increasing number of cases. As a result, diagnostic accuracy and disease classification have improved, while also paving the way for the potential development of targeted therapeutic strategies.

NEPHROTIC SYNDROME

Proteinuria in NS results from dysfunction of podocytes, leading to loss of the selective filtering function and increased permeability of glomerular filtration barrier. For detailed discussion on glomerular cell biology and podocytopathies, see [Chapter 4](#). The NS results from ongoing, substantive losses of protein, principally albumin, into the urine. The resulting characteristic features include urine protein excretion >3.5 g/day or a urine protein-to-creatinine ratio (uPCR) >3.5 g/g (>1 g/m² daily; >40 mg/m²/h in children), termed “nephrotic-range proteinuria”; hypoalbuminemia; hypercholesterolemia; and edema.

Although albumin is the predominant protein lost in NS, other macromolecules are also filtered abnormally, disrupting homeostasis in multiple physiologic systems ([Box 30.1](#)). This can affect immune function, coagulation, endocrine pathways, metabolism, and hematopoiesis, contributing to a wide range of complications. Among these, infections, acute kidney injury (AKI), and venous thromboembolism (VTE) are the most common and clinically significant, often requiring urgent management.

COMPLICATIONS OF NEPHROTIC SYNDROME**EDEMA**

Edema is the primary clinical symptom of NS and frequently prompts patients to seek medical attention. It results from sodium and water retention in the extravascular space and can range in severity from mild swelling to anasarca. Severe cases may be associated with transudative effusions including ascites, pleural effusions, or, more rarely, pericardial effusions. The mechanisms underlying edema formation have been debated and can be explained by two theories: the “overfill” and “underfill” hypotheses¹ (for further details, see [Chapter 29](#)). Briefly, the overfill hypothesis, now recognized as the predominant mechanism in most cases, suggests that primary sodium retention occurs due to overactivation of epithelial

sodium channels (ENaC) in the collecting duct.^{2,5} This sodium retention leads to water reabsorption, suppression of the renin-angiotensin-aldosterone (RAA) system and antidiuretic hormone (ADH), and increased release of natriuretic peptides. Blood volume generally remains normal or slightly elevated, though hypervolemia and hypertension can occur in some cases. In contrast, the underfill hypothesis attributes edema formation to hypoalbuminemia, which reduces plasma oncotic pressure and causes fluid to shift into the extravascular space, resulting in hypovolemia. This activates the sympathetic nervous system, the RAA system, and ADH, further promoting sodium and water retention and exacerbating edema. While the underfill mechanism lacks strong mechanistic support,^{6,7} it may be relevant in severe cases, particularly in children.^{8,9}

HYPOVOLEMIA

Hypovolemia needs to be excluded in all children presenting with NS, with or without edema, particularly in a setting of vomiting, diarrhea, reduced oral intake, or unsupervised diuretic use.^{10,11} Clinical features include history of abdominal pain, vomiting, or postural hypotension; lethargy; prolonged capillary refill time; cold extremities; tachycardia; low-volume pulses; and hypotension. Biochemical parameters that indicate hypovolemia include elevated hematocrit, high blood urea (mg/dL)-to-creatinine (mg/dL) ratio, low fractional excretion of sodium (<0.5%), and high urinary potassium index (>0.6).^{10,11} In children, dehydration and subsequent hypovolemia, often triggered by infectious complications, are major contributors to AKI. In adults, the mechanisms underlying AKI differ significantly. In MCD, AKI occurs in up to one-third of cases, and its leading cause is acute tubular necrosis (ATN), typically driven by tubular ischemia.¹²⁻¹⁴ In FSGS, AKI is less common but may occur in its collapsing variant, frequently linked to secondary causes such as viral infections or drug toxicity. MN rarely results in AKI; when it does, an alternative or overlapping etiology should be considered.

HYPERLIPIDEMIA

The severity of proteinuria in NS is closely linked to alterations in lipid metabolism, which involve both reduced clearance (the primary mechanism of hypercholesterolemia) and altered biosynthesis (the predominant mechanism of hypertriglyceridemia), as well as changes in lipid composition. This leads to markedly elevated levels of total cholesterol, low-density lipoprotein (LDL), triglycerides, and apolipoprotein B (ApoB)-containing lipoproteins (such as very low-density lipoprotein [VLDL], intermediate-density lipoprotein [IDL], and lipoprotein[a]). While high-density lipoprotein (HDL) levels typically remain unchanged, the lipoprotein itself undergoes impaired maturation and diminished cholesterol efflux function.¹⁵ For a detailed discussion, see [Chapter 29](#).

As a result, patients with NS exhibit a proatherogenic lipid profile, which accelerates the development of atherosclerosis. This leads to a 10-year cardiovascular event risk of 12.1% to 23.7%, approximately 2.5 times higher than in the general population, with the highest risk observed in FSGS, followed by MN.^{16,17} The cardiovascular risk in NS mirrors that seen in individuals with ESKF¹⁸ and is associated with traditional risk factors,¹⁹ as well as the degree of proteinuria and renal

function.^{20,21} Studies on FSGS have shown that proteinuria >1.5 g/day increases the risk of adverse cardiovascular events and all-cause mortality by 2.11-fold,²¹ and a urine albumin-to-creatinine ratio (uACR) >0.7 g/g results in a 3.37-fold higher risk.²⁰ Furthermore, hyperlipidemia contributes to the progression of chronic kidney disease (CKD) by exacerbating glomerular injury through lipid accumulation and oxidation, production of reactive oxygen species, recruitment of monocytes and macrophages, and subclinical inflammation. This leads to further glomerulosclerosis (the “lipid nephrotoxicity” hypothesis), in addition to promoting atherosclerosis of the renal vasculature. Lastly, elevated oxidized LDL levels increase the risk of thromboembolic events by enhancing platelet activation.¹⁵

VENOUS THROMBOEMBOLISM

Venous thromboembolism (VTE) is a potentially life-threatening complication, resulting from hypercoagulable state in NS. Hypercoagulability is a consequence of both increased platelet activity (and in some cases also increased platelet count) and altered plasma hemostatic factors activity. In addition to albuminuria, patients with NS lose low-molecular-weight anticoagulants (e.g., antithrombin III and protein S) and fibrinolytics (e.g., plasminogen, tissue plasminogen activator [t-PA]) in their urine. Hepatic overproduction of proteins in response to hypoalbuminemia leads to increased levels of high-molecular-weight coagulation factors (e.g., factor V, factor VIII, fibrinogen, and von Willebrand factor) and inhibitors of fibrinolysis (plasminogen activator inhibitor-1 [PAI-1], α_2 -macroglobulin). This predisposes patients to VTE, occurring in both typical sites (deep vein thrombosis of lower limbs, pulmonary embolism) and atypical locations (e.g., renal vein^{22,23} and dural sinuses²⁴).

The risk of VTE is highest during the early course of NS, with most cases occurring within the first 6 months of diagnosis and many detected simultaneously with the underlying glomerular disease.^{25,26} Data from Danish National Registry have estimated an 18-fold higher incidence of VTE within the first 30 days post diagnosis and a more than 7-fold higher incidence in the first year, compared with the general population.²⁷

This risk correlates strongly with the severity of NS, with hypoalbuminemia below 2.5 g/dL being the most reliable predictor.^{25,28,29} Among glomerular diseases, MN is associated with the highest VTE risk, including a predilection for renal vein thrombosis, though the mechanisms underlying this association remain unclear. The overall prevalence of VTE in MN is estimated at 7% to 8%,^{25,28} rising to 22% to 37% in patients with NS.^{23,30} Thromboembolic complications are also observed in MCD, occurring in 12% to 24% of adults^{23,31} but only 3% of children.²⁴ However, congenital NS (affecting children aged 0–3 months) carries an elevated VTE risk of up to 10%. Of note, VTE events occur more often in children developing severe infections during NS.²⁴

INFECTIONS

In the preantibiotic and preimmunosuppression era, infectious complications were the leading cause of death in children with NS, with mortality rates reaching up to 30%³²

and infections accounting for 60% of these cases.³³ Advances in medical care have nearly eliminated fatal infections, yet severe infections remain a significant burden, affecting approximately 40% of children²⁴ and around 10% to 15% of adults with NS.^{31,34,35} The mechanisms predisposing patients with NS to infections are multifaceted, involving impairments across all branches of the immune system. Notably, urinary losses of immunoglobulin G (IgG) and components of the alternative complement pathway (e.g., factors B and I) contribute to reduced humoral and innate immunity. Additionally, altered T-cell function exacerbates immune deficiencies. These immunologic changes align with the historically observed susceptibility to infections by encapsulated bacteria, which rely on complement-mediated opsonization for elimination.³⁶⁻⁴⁰ *Streptococcus pneumoniae* remains the most frequently implicated pathogen in these infections.²⁴ Local factors also play a role in the development of infectious complications. These include fluid accumulation, which predisposes to peritonitis (the most commonly reported infection in children)²⁴ and cellulitis. Other contributing factors are hypercatabolism, malnutrition, and the use of immunosuppressive medications, which further compromise immune defenses.

METABOLIC AND ENDOCRINE DYSFUNCTION

Malnutrition and muscle mass loss, although not systematically studied, may complicate the course of NS as a result of hypercatabolism. Similarly, disturbances in hormonal and microelement homeostasis may arise from the urinary loss of carrier proteins, along with bound fraction of hormones. Thyroid dysfunction is the most common hormonal abnormality in NS,⁴¹ with hypothyroidism predominating in children,⁴² while both hypothyroidism and sick euthyroid syndrome occur at similar frequencies in adults.⁴³ Other potential disturbances include deficiencies in growth hormone, sex hormones, vitamin D, and iron, among others.⁴¹ The clinical implications of these hormonal and metabolic abnormalities remain uncertain, particularly in cases responding to therapy where they are transient. However, persistent NS warrants assessment and management of these derangements to mitigate potential long-term complications.³⁹

MANAGEMENT OF PATIENTS WITH NEPHROTIC SYNDROME

Management of NS includes three key components: supportive therapy, applicable to all patients; immunosuppressive therapy, reserved for those with immune-mediated disease; and chemoprophylaxis, used to mitigate the side effects of specific immunosuppressive medications. The details of these treatment approaches are outlined in Fig. 30.2.

CONSERVATIVE MANAGEMENT

Supportive therapy plays an important role in management of patients with NS. It consists of lifestyle modification (diet, physical activity) aiming to reduce long-term cardiovascular risk and slow the progression of CKD; pharmacologic nephroprotective strategies (including therapy with

angiotensin-converting enzyme inhibitors [ACEi] or angiotensin receptor blockers [ARB] and potentially sodium-glucose cotransporter-2 inhibitors [SGLT2i] as an add-on therapy); management of symptoms related to NS (e.g., lipid-lowering therapy for hyperlipidemia and diuretics for edema) and preventing its acute complications (thromboprophylaxis, vaccination); and proper and effective therapy of comorbid conditions (e.g., hypertension and diabetes).³⁷⁻⁴⁰

Lifestyle recommendations for NS generally align with those for the healthy population. Sodium intake should be limited to 2 g/day (equivalent to 5 mg/day or 90 mmol of sodium chloride), while protein intake should be maintained at 0.8 to 1.0 g/kg/day, with additional protein equivalent to daily proteinuria (up to a maximum of 5 g/day) if kidney function is normal. In cases of impaired kidney function, protein intake should be reduced to 0.8 g/kg/day.⁴⁰ High sodium and protein intake can lead to glomerular hyperfiltration, accelerating CKD progression and increasing proteinuria. Given that NS is a hypercatabolic state, energy intake should be increased to 35 kcal/kg/day. A heart-healthy diet is recommended, with total fat intake limited to <30%, saturated fat to <7% to 10%, and cholesterol to <200 mg/day. Patients should maintain a healthy body weight, engage in regular physical activity, and avoid smoking.

The cornerstone of conservative management is the initiation and titration of medications that block the renin-angiotensin-aldosterone system (RAAS), such as ACEi or ARB, to the maximum tolerated dose. The combination of these medications is generally not recommended. These drugs have demonstrated clear benefits in improving kidney outcomes by slowing the decline in estimated glomerular filtration rate (eGFR), reducing proteinuria, and enhancing the likelihood of spontaneous remission in MN.⁴⁰ Their cardioprotective effects are also well established.

Sodium-glucose cotransporter 2 inhibitors (SGLT2i), known for improving renal and cardiovascular outcomes in CKD,^{44,45} should be considered as an adjunctive therapy after reaching the maximum tolerated dose of ACEi/ARB. Robust data support the use of SGLT2i in nondiabetic patients with CKD, and emerging evidence from ancillary analyses of clinical trials in glomerular diseases shows similar benefits.⁴⁶ However, patients on immunosuppressive therapy were excluded from these studies.^{47,48} A retrospective study has demonstrated a positive effect of SGLT2i on proteinuria in patients with various glomerular diseases, although it was underpowered to detect effects on eGFR.⁴⁹

The metabolic effects of SGLT2i, such as modest reductions in body weight and blood pressure, lower serum glucose and uric acid, prevention of magnesium loss in urine, and increased natriuresis, offer potential benefits for patients receiving immunosuppressive therapies. For instance, glucocorticoids can cause weight gain, hypertension, hyperglycemia, and hyperuricemia, and calcineurin inhibitors (CNIs) can lead to hypomagnesemia, hyperuricemia, hyperglycemia, and hypertension, which may be mitigated by SGLT2i. Additionally, SGLT2i may help manage edema.

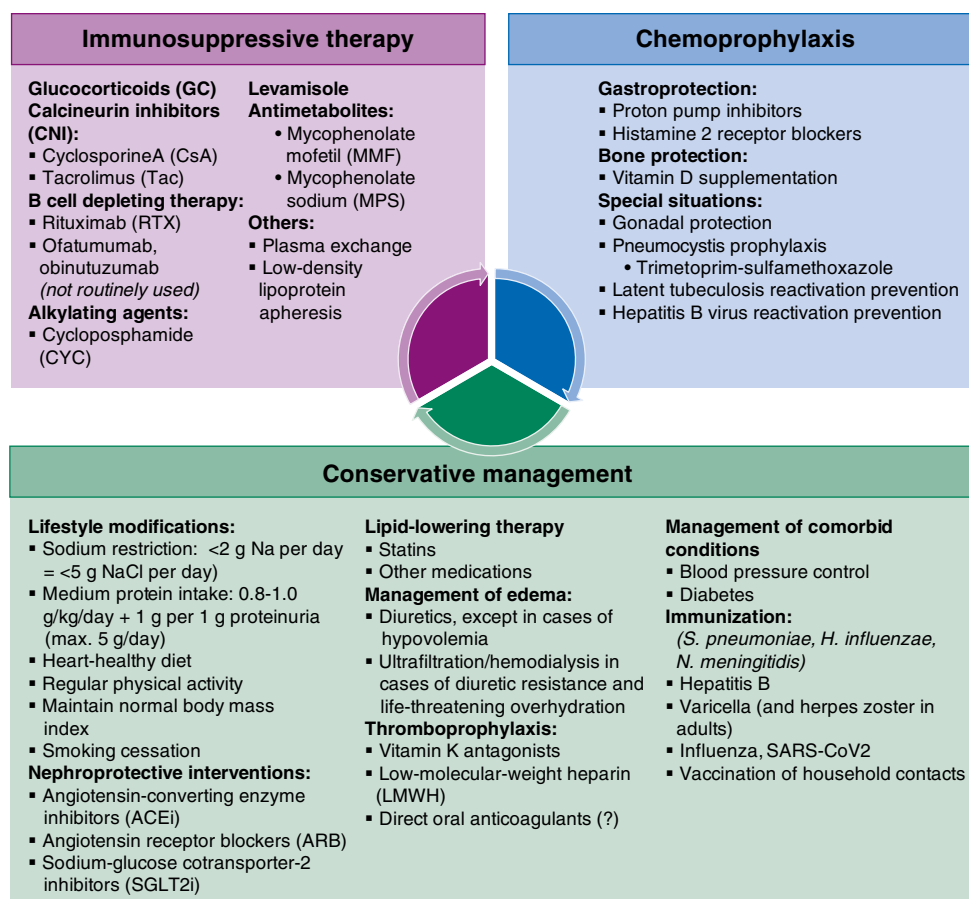


Fig. 30.2 Overview of management of patients with nephrotic syndrome.

Nephrocardioprotective therapy is recommended for all patients except children with steroid-sensitive NS (SSNS) and adults with MCD who have preserved kidney function and normal blood pressure.^{38,39,50} An important component of nephrocardioprotection includes optimal management of comorbidities, particularly blood pressure control, with a target systolic blood pressure of <120 mm Hg (or ≤50th percentile in children), if tolerated, or at least 120 to 130 mm Hg.^{40,51}

Lipid-lowering therapy is also important. It is recommended to treat hyperlipidemia in steroid-resistant NS (SRNS) in children and nonremitting NS cases in adults following guidelines for general population regarding the assessment of cardiovascular risk, target LDL values, and therapeutic options. Of note, hyperlipidemia in NS is usually severe and achievement of target levels may be challenging, requiring combined therapy. Statins and proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors may be particularly beneficial as they directly target lipoprotein metabolism pathways that are altered in NS.^{15,52,53} Statins are considered the first-line medication, while the real-world evidence of efficacy and safety of PCSK9 inhibitors in NS is limited thus far. Patients with MN with hypoalbuminemia of 2.5 to 3.2 and high cardiovascular risk should be additionally offered acetylsalicylic acid (ASA).

Edema is the hallmark clinical feature of NS, and the most effective therapy is achievement of remission of the disease.

Symptomatic treatment includes dietary salt restriction and diuretic use, except in cases of concomitant hypovolemia. NS, particularly severe, is considered related to “diuretic resistance”—a state requiring higher drug doses, switching to intravenous route of administration, and/or combination of drugs of different targets to achieve the adequate response.⁵⁴ This is due to edema of the gut mucosa, resulting in altered drug bioavailability; hypoalbuminemia affecting volume of distribution and delivery to target ionic channels, as well as increased renal clearance secondary to podocyte injury; and increased permeability of the glomerular filtration barrier.^{55,56} The rationale for using intravenous 20% albumin solution remains controversial. It has been shown that the natriuretic effect of albumin equals the sodium load provided as the drip solvent, arguing against any diuretic benefit. The accompanying significant albumin load is immediately leaked into the urine, aggravating proteinuria and its nephrotoxic effects.⁵⁷⁻⁵⁹ Intravenous albumin may therefore be considered only as a “salvage therapy” in cases of complete diuretic resistance before ultrafiltration (either isolated or during hemodialysis) may be required. In contrast to edema management, intravenous albumin administration is justifiable and recommended for hypovolemia in the course of NS, observed particularly in children^{38,39} (Fig. 30.3).

An integral component of conservative therapy in severe cases of NS is prophylaxis of VTE events. The high-quality

data on the optimal approach—indications, drug choice, and duration of thromboprophylaxis—are, however, lacking. Prophylaxis with warfarin or low-molecular-weight heparin (LMWH) is recommended in patients with MN, showing the most prominent association with the risk of VTE, with severe NS with serum albumin <2.5 g/dL and concomitant low risk of bleeding; otherwise, acetylsalicylic acid should be considered.^{40,60} A potential benefit-risk ratio in MN may be calculated using the GN tools algorithm (<https://www.med.unc.edu/gntools/>).²⁹ In other glomerular diseases manifesting as NS, no direct guidelines are provided. The suggested indications for thromboprophylaxis include serum albumin of <2.0 to 2.5 g/dL and additional risk factors, such as massive proteinuria (>10 g/day), obesity (II–III degrees: body mass index >35 kg/m²), genetic mutations (or family history) predisposing to VTE, severe heart failure, recent surgery or prolonged immobilization, and the absence of contraindications to anticoagulation.⁴⁰ Existing evidence has suggested clinical efficacy of vitamin K antagonists and LMWH^{29,61}; small studies have also reported efficacy and safety of direct oral anticoagulants (DOACs, such as apixaban, rivaroxaban, dabigatran).^{62,63} However, it has been emphasized that DOACs' effect may be diminished in hypoalbuminemia due to moderate-to-high albumin binding, while heparin effect may be affected by antithrombin deficiency (being the target for heparin).⁴⁰ In-depth pharmacokinetic studies in this regard are not available for any of these groups, but small studies have reported reduced anti-Xa activity of LMWH⁶⁴ with the need for doubled doses to achieve target levels.⁶⁵ This, however, seems to not affect clinical efficacy of standard LMWH dosing, shown to be successful in preventing VTE in a cohort of 143 patients.⁶¹ Despite limited data, one survey conducted among nephrologists has revealed approximately half of them prescribing DOACs,⁶⁶ presumably as the most convenient to use, and known to have the lowest risk of bleeding complications in the general population.

Morbidity due to severe infections, associated with disease relapses and immunosuppression, has declined with prompt diagnosis and immunization practices; however, they remain the chief cause of hospitalization.⁶⁷⁻⁷⁰ This underscores the importance of vaccination as a key preventive health measure.⁷¹ At disease onset, patients' vaccination status should be reviewed and updated, ensuring completion of all vaccines according to national guidelines. Priority should be given to vaccines targeting encapsulated bacteria (*Streptococcus pneumoniae*, *Neisseria meningitidis*, and *Haemophilus influenzae*); hepatitis B; and varicella in children (live vaccine) or herpes zoster (recombinant vaccine) in adults. Annual influenza vaccination and repeating COVID-19 vaccination in line with national recommendations are also advised. The live-attenuated varicella vaccine is particularly important in children, as chickenpox can have a life-threatening course in immunocompromised young individuals. However, live vaccines (including those for varicella, measles, mumps, rubella, rotavirus, and yellow fever) are contraindicated during immunosuppressive therapy. Ideally, these vaccines should be administered at least one month before initiating immunosuppression. If this is not feasible, live vaccination should be deferred until at least 6 to 9 months after rituximab (RTX) dose, with confirmed B-cell reconstitution, and until

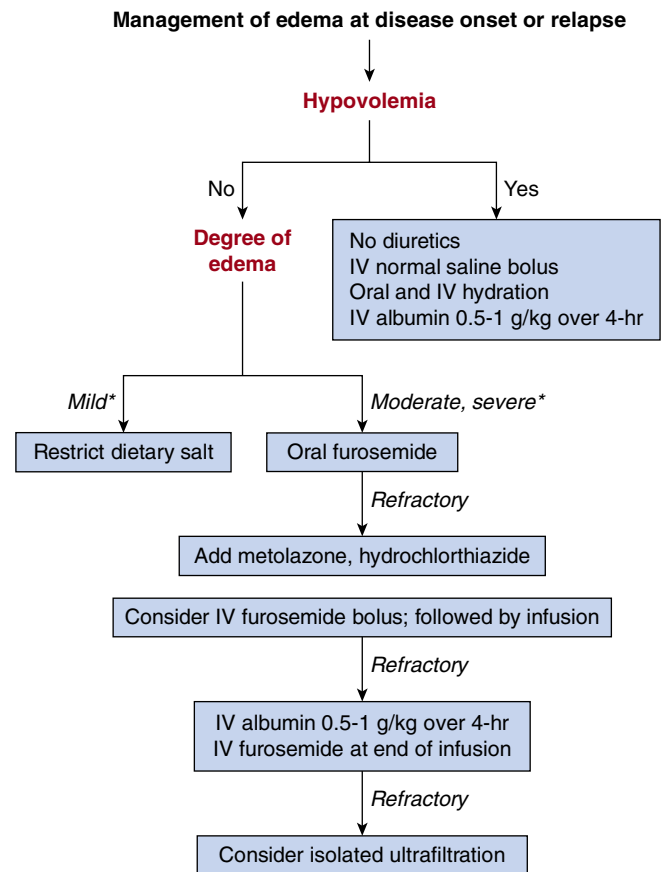


Fig. 30.3 Management of edema and hypovolemia in nephrotic syndrome. *In children, edema severity is graded empirically as follows: mild ($\leq 7\%$ increase in body weight), moderate (7% to 15%), and severe ($>15\%$). No grading system has been established for adult patients. (Adapted from Sinha A, Bagga A, Banerjee S, et al. Steroid sensitive nephrotic syndrome: revised guidelines. *Indian Pediatr.* 2021;58:461–481.)

other immunosuppressive agents have been discontinued for at least 1 to 3 months. This minimizes the risk of live vaccine-induced infections, which, however, is generally low according to current evidence.⁴⁰ Inactivated vaccines can be administered safely during immunosuppressive therapy but may show reduced efficacy, particularly with prednisone doses >20 mg/day for more than 2 weeks or during RTX treatment.⁷² For optimal seroconversion, administration timing should follow the same guidelines as live vaccines where possible. Vaccination of household contacts is also recommended as part of a comprehensive preventive strategy.^{38,39}

IMMUNOSUPPRESSIVE THERAPY

The immunosuppressive therapies used in treating NS target various mechanisms of the immune response, reflecting the complexity of the disease's underlying pathophysiology.

Glucocorticoids act by binding to glucocorticoid receptors, which modulate gene expression and, to a lesser extent, post-translational pathways. This suppresses the production of proinflammatory cytokines, inhibits immune cell activation, and reduces T-cell proliferation, effectively dampening inflammation.

CNIs, including cyclosporine (which binds cyclophilin) and tacrolimus (which binds FK-binding proteins), inhibit calcineurin, a phosphatase necessary for activating nuclear factor of activated T cells (NFAT). This inhibition blocks T-cell activation and the production of interleukin-2 (IL-2), a critical mediator of T-cell proliferation. CNIs primarily target T-helper cells but also exhibit antiproteinuric effects. These include direct hemodynamic actions, such as afferent arteriole vasoconstriction, and cytoskeletal stabilization of podocytes. CNIs prevent the dephosphorylation of synaptopodin, an actin-binding protein essential for maintaining podocyte structure and preventing FPE.

Mycophenolate acid derivatives, such as mycophenolate mofetil (MMF) and mycophenolate sodium (MPS), inhibit inosine monophosphate dehydrogenase, a key enzyme in de novo purine synthesis. This action preferentially reduces the proliferation of T and B lymphocytes, which are dependent on this pathway for nucleotide synthesis.

Cyclophosphamide, an alkylating agent, induces DNA cross-linking, leading to apoptosis, particularly in rapidly dividing immune cells. This broadly suppresses both B and T-cell activity, contributing to its immunosuppressive effects.

Rituximab, a monoclonal anti-CD20 antibody, targets B cells, causing their depletion. Beyond its effects on B-cell immunity, rituximab may stabilize the podocyte actin cytoskeleton by increasing the expression of sphingomyelin phosphodiesterase acid-like 3b (SMPDL3b), a protein involved in cytoskeletal integrity. Notably, it is the only biologic routinely used for NS treatment.

This broad spectrum of immunosuppressive mechanisms reflects the multifaceted approach necessary for managing the diverse etiologies and pathophysiology of NS.

CHEMOPROPHYLAXIS

The final component of conservative management involves mitigating the side effects associated with immunosuppressive therapies. Key measures include gastrointestinal and bone protection during glucocorticoid therapy, gonadal protection when using cyclophosphamide, and, importantly, prevention of latent infections such as tuberculosis, *Pneumocystis jiroveci*, and hepatitis B reactivation during intensive immunosuppression, following local clinical guidelines.

ETIOLOGY OF NEPHROTIC SYNDROME

The etiology and patterns of injury leading to the clinical manifestations of NS vary across the lifespan,^{33,73} as illustrated in Fig. 30.4, and may also differ by ethnicity.⁷⁴ In general, as previously mentioned, the primary causes of NS are podocytopathies, which exhibit one of three main patterns of injury: MCD, FSGS, and MN, which are discussed in detail later. Additionally, NS can present with other patterns of injury: amyloidosis (discussed in Chapters 35, 44, and 59 briefly summarized below) and diabetic kidney disease (see Chapter 41). Rarely, NS may be associated with other conditions, such as fibrillary glomerulonephritis or immunotactoid glomerulonephritis (see below). Finally, NS can overlap with nephritic syndrome in the context of various forms

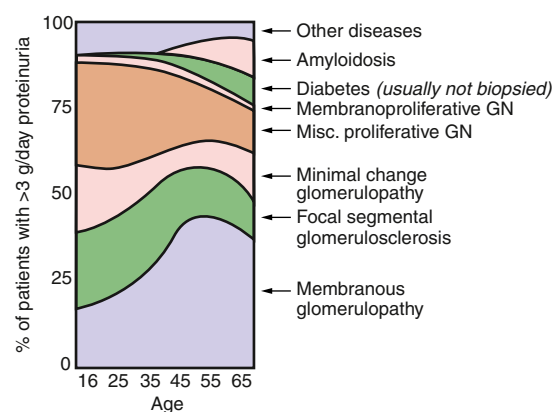


Fig. 30.4 Graphic depiction of frequencies of different forms of glomerular disease identified in kidney biopsy specimens from patients with proteinuria of more than 3 g of protein/day evaluated at the University of North Carolina Nephropathology Laboratory. Some diseases that cause proteinuria are underrepresented because they are not always evaluated by kidney biopsy. For example, in many patients, steroid-responsive proteinuria is given a presumptive diagnosis of minimal change disease and patients do not undergo biopsy, and most patients with diabetes and proteinuria are presumed to have diabetic glomerulosclerosis and do not undergo biopsy. GN, Glomerulonephritis.

of glomerulonephritis, which are discussed in other chapters (see Chapters 31–36).

MINIMAL CHANGE DISEASE

Minimal change disease (MCD) is a pattern of injury named for the normal appearance of glomeruli on light microscopy and immunofluorescence, with histologic abnormalities detectable only on electron microscopy, showing diffuse podocyte FPE. It is the leading cause of NS in children, accounting for 70% to 90% of cases. In pediatric populations, kidney biopsies are typically not performed, and such cases are classified as “idiopathic nephrotic syndrome” (INS). Studies have identified MCD as an autoimmune disease mediated by autoantibodies targeting slit-diaphragm antigens, particularly nephrin in a substantial proportion of cases. The clinical course is characterized by an abrupt onset of NS and an equally rapid, complete remission (CR) with glucocorticoid (GC) therapy (SSNS) in >80% to 90% of patients. MCD/INS has an excellent long-term prognosis, with <5% of patients developing ESKF. Nevertheless, >70% experience relapses, necessitating repeated courses of immunosuppressive therapy. Preventing and managing relapses while minimizing GC exposure and associated complications remain a key clinical challenge.

EPIDEMIOLOGY

MCD can occur at any age but is most prevalent in childhood,⁷⁵ where it is classified as INS in children managed without a kidney biopsy. INS is a rare disease with a prevalence of 12 to 16 per 100,000 children,^{32,76} accounting for 70% to 90% of NS cases in children younger than 10 years of age (see Fig. 30.4). The average incidence of INS is 2.92 per 100,000

children per year, with higher rates in Southeast and East Asia compared with Europe, North America, and Oceania.³² Peak incidence occurs between ages 2 and 6, with a male-to-female ratio of 2–3:1⁷⁷ and increased prevalence in children with atopic diseases, observed four times more often in INS than in the general population.⁷⁸ In adults, MCD constitutes 10% to 20% of histopathologic diagnoses,⁷⁹ with an incidence of 0.6 per 100,000 adults per year,⁸⁰ particularly affecting young adults and the elderly, where it may be associated with malignancies.

ETIOLOGY AND PATHOPHYSIOLOGY

Primary MCD is a kidney-limited disease. Secondary forms of MCD are uncommon (eBox 30.1) and may result from drug toxicity (e.g., nonsteroidal antiinflammatory drugs (NSAIDs), bisphosphonates, lithium, penicillins, and rifampicin); systemic lupus erythematosus (SLE)-termed lupus podocytopathy (see Chapter 31)⁸¹; or malignancies, especially hematologic disorders.⁹

Primary MCD, as well as INS and primary FSGS, are associated with the loss of immunologic tolerance to the podocyte antigen(s) and the production of autoantibodies against the target antigen(s). The “two-hit theory” of the development of autoimmune diseases posits that environmental triggers may initiate the disease in genetically susceptible individuals—with polygenic predisposition associated with dysregulation of the immune system. Indeed, genome-wide association studies, including multiple international cohorts, have identified polymorphisms in human leukocyte antigens (HLA), specifically HLA-DR and HLA-DQ, associated with INS susceptibility, as well as polymorphisms in various genes regulating immune responses.^{82–86a} The approximately fourfold higher prevalence of atopic diseases in individuals with SSNS, especially with early onset, further supports the role of immune system abnormalities in disease development.⁷⁸ No specific triggering factors have been definitively identified; however, potential triggers include infections, particularly viral upper respiratory tract infections, explaining the seasonal incidence with peaks in winter⁸⁷; and exposure to allergens.⁸⁸

Discoveries have advanced our understanding of the immunologic mechanisms leading to podocyte injury and altered glomerular permeability in INS, MCD, and primary FSGS. Since 1954, when nephrotic serum from INS patients was shown to induce proteinuria in healthy individuals, efforts have focused on identifying the so-called “circulating permeability factor.”⁸⁹ This hypothesis has been supported by the recurrence of nephrotic-range proteinuria almost immediately post transplant in 30% to 60% of patients with nongenetic FSGS,^{90–92} successful retransplantation of such a kidney to another recipient leading to resolution of proteinuria,⁹³ and the efficacy of plasma exchange in preventing FSGS recurrence after transplantation.⁹⁴ Clinical evidence has been further supported by experimental studies indicating that nephrotic plasma alters podocyte cytoskeletal distribution, induces FPE, and increases glomerular permeability.⁹⁵

For the past 50 years, INS, MCD, and FSGS have been considered T cell-mediated diseases, prompting investigations into associated mediators.⁹⁶ Proposed circulating factors include interleukin 13

(IL-13), cardiotrophin-like cytokine factor 1 (CLCF-1), tumor necrosis factor- α (TNF- α), hemopexin, heparinase, c-maf inducing protein, soluble urokinase-type plasminogen activator receptor (suPAR), vascular endothelial growth factor (VEGF), anti-CD40 antibody, and soluble CD40 ligand (sCD40L).^{97–99} Other implicated factors include angiopoietin-like 4 (ANGPTL4)¹⁰⁰ and circulating proteases.¹⁰¹ However, studies on these candidate proteins have yielded inconsistent results, failing to unequivocally confirm their pathogenic roles.

Emerging evidence highlights the role of B cells in disease pathogenesis, as demonstrated by the efficacy of anti-CD20 therapies (e.g., rituximab^{102,103} and ofatumumab¹⁰⁴) and the association of reconstituted switched memory B cells with the risk of NS relapse. In 2022, a breakthrough discovery using high-resolution confocal microscopy revealed punctate IgG deposits colocalizing with nephrin, a key slit-diaphragm protein, and demonstrated the presence of nephrin antibodies in the serum of patients with biopsy-proven MCD, establishing the antibody-mediated nature of the disease.¹⁰⁵ Further study employing a sensitive hybrid technique that combines immunoprecipitation with enzyme-linked immunosorbent assay (ELISA) has identified nephrin antibodies in the majority of treatment-naïve patients with ongoing NS, with 90% of children with INS and 79% of adults with MCD testing positive.¹⁰⁶ Additionally, mice model study has confirmed the pathogenicity of nephrin antibodies, representing a significant advancement in our understanding of the disease’s pathogenesis.¹⁰⁶ The identification of punctate IgG deposits colocalizing with nephrin in 31% to 100% of patients with recurrent FSGS following transplantation,^{107,108} along with the detection of corresponding antibodies in 75% of pretransplant sera from patients experiencing disease recurrence¹⁰⁷ and 100% at the time of recurrence,¹⁰⁸ supports the hypothesis of a shared pathogenic mechanism among primary podocytopathies, including INS, MCD, and FSGS. This evidence supports the thesis that primary (immune-mediated) FSGS may represent a histologic progression of MCD.

Positive punctate IgG staining has also identified a subset of patients with presumed primary MCD or FSGS among steroid-resistant cases, specifically those who respond to intensified immunosuppressive therapy and achieve favorable long-term renal outcomes with minimal risk of progression to ESKF. Most of these cases exhibit secondary steroid resistance rather than primary (87.5% vs. 27%), with 77% demonstrating IgG deposits colocalizing with nephrin. Notably, approximately 23% of SRNS cases with punctate IgG deposits on confocal microscopy lack nephrin colocalization, suggesting the involvement of alternative antislit-diaphragm antibodies.¹⁰⁹ Indeed, some reports indicate that sera from these patients contain antibodies directed against other podocyte antigens, including ubiquitin carboxyl-terminal hydrolase L1 (UCHL1),¹¹⁰ annexin A2,¹¹¹ and other,¹¹² though these findings require further validation and confirmation of pathogenic roles of these antibodies.

The identification of nephrin as a target antigen, along with its corresponding autoantibodies, has transformed the understanding of this group of podocytopathies, reclassifying the disease as antibody mediated. This shift is likely to prompt updates in disease classification and

eBox 30.1 Common Associations With Minimal Change Disease**Infections**

- Viral
- Parasitic

Pharmaceutical Agents

- Nonsteroidal antiinflammatory drugs
- Gold
- Lithium
- Interferon
- Ampicillin
- Rifampin
- Trimethadione
- Tiopronin

Tumors

- Hodgkin disease
- Lymphoma, leukemia
- Solid tumors

Allergies

- Food
- Dust
- Bee stings
- Pollen
- Poison ivy and poison oak
- Dermatitis herpetiformis

Disease and Other Associations

- Systemic lupus erythematosus
- Following allogeneic stem cell transplantation for leukemia
- Following hematopoietic cell transplantation

Data from references. [128,375,376,377,378](#)

nomenclature in the near future. The considerable variation in the prevalence of these autoantibodies highlights the necessity for optimizing and validating diagnostic methods before their commercial application. Additionally, future research may uncover further podocyte-specific target antigens in certain patient subgroups.

PATHOLOGY

In MCD, light microscopy typically reveals no glomerular lesions or only mild mesangial prominence, while immunofluorescence is negative or shows only trace, nonspecific staining for IgM and/or C3 (Figs. 30.5A and

30.6A). Electron microscopy confirms diffuse podocyte FPE without the presence of electron-dense deposits (Figs. 30.5B and 30.6B). It is important to note that early-stage FSGS may be misdiagnosed as MCD if the biopsy sample is inadequate.

CLINICAL PRESENTATION AND DISEASE COURSE

PRIMARY MINIMAL CHANGE DISEASE

Clinical picture

The hallmark clinical feature of MCD is the sudden onset of nephrotic syndrome.¹¹³ The initial episode usually occurs in children aged 2 to 6 years and is often preceded by an

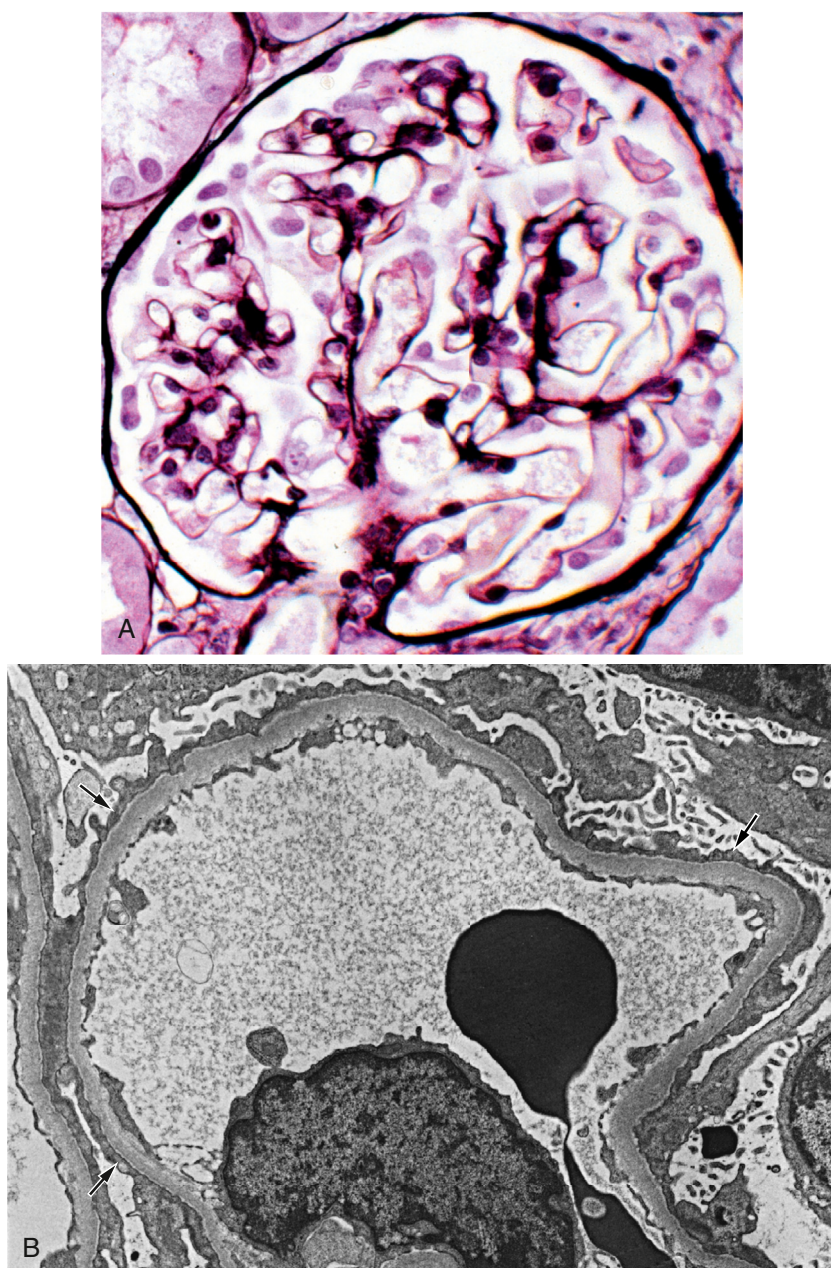


Fig. 30.5 (A) Unremarkable light microscopy appearance of a biopsy specimen from a patient with minimal change disease. Glomerular basement membranes are thin, and there is no glomerular hypercellularity or mesangial matrix expansion. (Jones methenamine silver stain, $\times 300$.) (B) Electron micrograph of a glomerular capillary wall from a patient with minimal change disease showing extensive podocyte foot process effacement (arrows) and microvillous transformation. ($\times 5000$.)

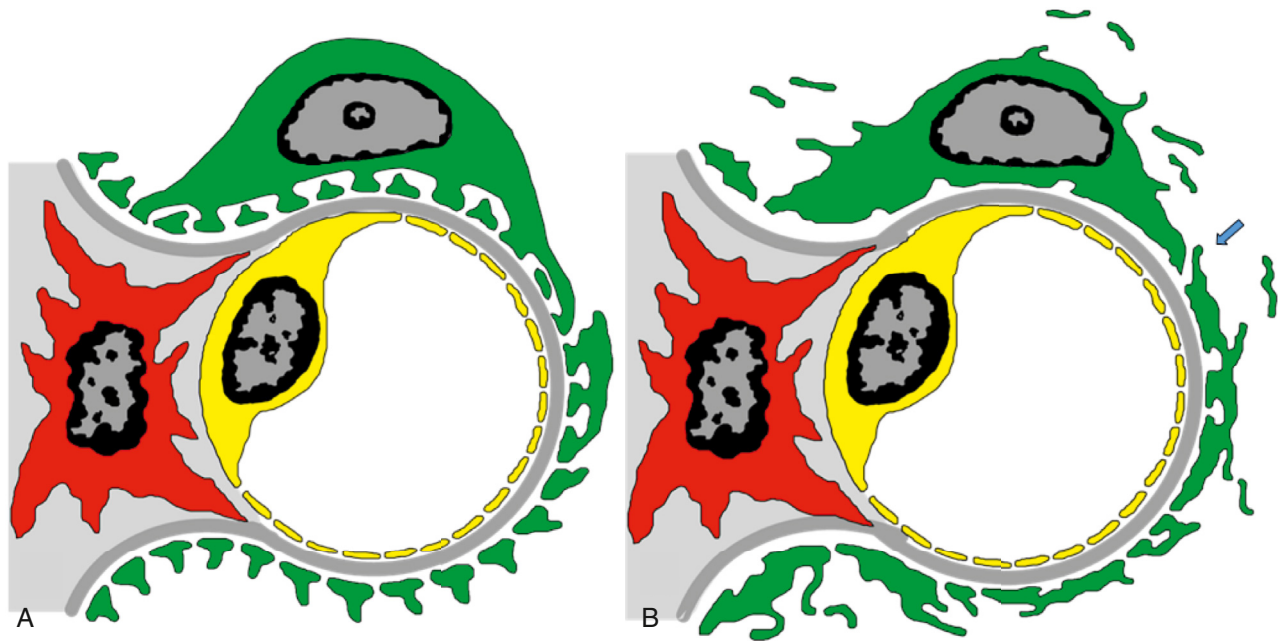


Fig. 30.6 Diagrammatic representations of the ultrastructural features of a normal glomerular capillary loop (A) and a capillary loop with features of minimal change disease (B). The latter has effacement of podocyte foot processes and microvillous projections of podocyte cytoplasm. (Courtesy J. C. Jennette.)

infection. Hematuria is uncommon but may occur in up to 10% to 24% of severe cases, most often in adults.^{31,114} Blood pressure is typically normal, and kidney function is preserved.

In adults, AKI has been observed in 20% to 37% of cases¹⁴ and appeared to result primarily from ischemic mechanisms, leading to ATN with endothelin-1 as a proposed mediator.^{12,13} Risk factors include older age, lower albumin levels, severe proteinuria, and hypertension.^{12,31,115,116} In children, AKI is less frequent (reported in 0.8%–24% of cases)¹¹⁷ and is more commonly associated with prerenal cause due to hypovolemia, with infectious complications being the most common predisposing factor.¹¹⁸ AKI episodes can worsen long-term renal outcomes.^{31,118}

Clinical course

Spontaneous remissions in MCD are rare, warranting immunosuppressive treatment in all cases. MCD typically responds rapidly and completely to high-dose glucocorticoids—a clinical course of SSNS. CR occurs in >90% of pediatric INS patients within 4 to 6 weeks of treatment and in >80% of adults, though typically more gradually—with only half of patients responding within 4 weeks and an additional 10% to 25% within weeks 5 to 16.^{116,119,120} It results in different SSNS definitions for children and adults, defined as CR within 6 or 16 weeks of high-dose glucocorticoids, respectively (Table 30.1).¹²¹

In children with INS, relapse rates are estimated at 71.9% (95% prediction interval: 38.8%–95.5%),³² with the majority occurring within the first year after disease onset—in approximately 44% of childhood-onset and 30% of adolescent-onset cases).¹²² About 50% of these patients experience infrequent relapses, while the remaining half require ongoing glucocorticoid therapy to control the disease (steroid-dependent nephrotic syndrome) or experience frequent relapses (frequently relapsing

nephrotic syndrome) (definitions provided in Table 30.1).³⁸ Frequent relapses and steroid dependence are more common in patients with early age at onset (<3 years), delayed time to initial remission, brief initial corticosteroid therapy, and short duration of initial remission.^{33,67,68} Relapse frequency tends to decrease with age (median time to relapse is 246 days in adults vs. 159 days in children)¹²²; however, 10% to 30% of patients with childhood disease onset continue to experience relapses in adulthood,³⁸ particularly those with early age at onset and frequently relapsing disease course.^{67,70} In adults, the hazard of relapse is 69% lower compared with children (hazard ratio [HR] for children vs. adults: 1.69, 95% confidence interval [CI]: 1.29–2.21).¹²² Most adults experience infrequent relapses, though up to one-third of patients present with steroid dependence or frequently relapsing courses of the disease.¹¹⁶

Approximately 10% to 20% of adult and 10% to 15% of pediatric patients demonstrate steroid resistance (SRNS), either at onset (primary SRNS) or following a subsequent relapse (secondary SRNS).¹¹⁶ Among primary SRNS cases in children, 45% achieve partial or CR with intensified immunosuppression, particularly CNIs, suggesting an immune-mediated cause of the disease. Genetic causes account for about 15% of remaining cases¹²³; however, in genetic NS, kidney biopsy more often reveals FSGS than MCD pattern (see “Focal Segmental Glomerulosclerosis” section later). In patients with no genetic mutations and no response to intensified immunosuppression, the etiology remains undetermined.

SECONDARY MINIMAL CHANGE DISEASE

Secondary forms of MCD are rare (eBox 30.1). Among malignancies, MCD is commonly associated with lymphoid malignancies, particularly Hodgkin lymphoma (0.4% of

Table 30.1 Definitions of Disease Course in Nephrotic Syndrome in Children and Minimal Change Disease and Focal Segmental Glomerulosclerosis in Adults

Term	Definition in Pediatric Nephrotic Syndrome	Definition in adult MCD/FSGS
Nephrotic syndrome	Edema, heavy proteinuria (>1 g/m ² per day; >40 mg/m ² /h), and hypoalbuminemia (serum albumin <3.0 g/dL)	Proteinuria >3.5 g/day or uPCR >3500 mg/g (or >350 mg/mmol), serum albumin <3.0 g/dL hypoalbuminemia, with or without edema
Complete remission	Urine protein nil-trace (dipstick) for 3 consecutive days, uPCR <0.2 mg/mg, or 24-h protein <100 mg/m ² /day	Proteinuria <0.3 g/day or uPCR <300 mg/g (or <30 mg/mmol), stable creatinine and serum albumin > 3.5 g/dL
Partial remission	Urine protein 1+/2+ (dipstick), uPCR 0.2-2.0 mg/mg, or 24-h urine protein 100-1000 mg/m ² /day; serum albumin ≥3.0 g/dL; and absence of edema	Proteinuria 0.3-3.5 g/day or uPCR 300-3500 mg/g (or 30-350 mg/mmol) and a decrease by >50% from baseline
Nonresponse	Urine protein 3+/4+ (dipstick), uPCR >2.0 mg/mg, or 24-h urine protein >1000 mg/m ² /day; albumin <3.0 g/dL; or edema	Persistence of proteinuria >3.5 g/day or uPCR >3500 mg/g (or >350 mg/mmol) with <50% reduction from baseline
Stable remission	Sustained remission or infrequent relapses during immunosuppressive therapy	Not specified
Relapse	Urine protein ≥3+ (uPCR >2 mg/mg) for 3 consecutive early morning specimens, having been in remission previously	Proteinuria >3.5 g/day or uPCR >3500 mg/g (or >350 mg/mmol) after complete remission
Frequent relapses	Two or more relapses in the first 6 months after stopping initial therapy; ≥3 relapses in any 12-months	≥2 relapses per 6 months or ≥4 relapses per 12 months
Steroid dependence	Two consecutive relapses when on alternate-day steroids or within 14 days of its discontinuation	Relapse during or within 2 weeks of discontinuation of glucocorticoid therapy
Difficult-to-treat steroid-sensitive disease	Both of the following: (i) frequent relapses, or significant steroid toxicity with infrequent relapses; and (ii) failure of ≥2 steroid-sparing agents (including levamisole, cyclophosphamide, mycophenolate mofetil)	Not specified
Significant steroid toxicity	Patients with hyperglycemia (fasting glucose >100 mg/dL, post prandial >140 mg/dL, or HbA1c >5.7%); obesity (body mass index >2 standard deviation scores for age and sex); short stature (<-2 SDS for age) with height velocity (<-3 SDS for age); raised intraocular pressure; cataract; myopathy; osteonecrosis; psychosis	Not specified
Steroid resistance	Lack of complete remission despite therapy with daily prednisone at a dose of 2 mg/kg (or 60 mg/m ²) daily for 6 weeks	Persistence of proteinuria >3.5 g/day or uPCR >3500 mg/g (or >350 mg/mmol) with <50% reduction from baseline despite therapy with daily prednisone at a dose of 1 mg/kg/day or 2 mg/kg every other day for >16 weeks
CNI-resistant disease	Nonresponse to cyclosporine A or tacrolimus, given in adequate doses and titrated to blood levels, for 6 months	No response to cyclosporine therapy at trough levels of 100-175 ng/mL or tacrolimus at trough levels of 5-10 ng/mL for 4-6 months
Monogenic disease	Pathogenic or likely pathogenic variation, defined by American College of Medical Genetics and Genomics, in a gene associated with nephrotic syndrome (see Table 30.2)	Not specified
Recurrence of nephrotic syndrome in allograft	Persistent proteinuria (uPCR >1) if previously anuric; or increase of uPCR by >1 mg/mg if proteinuria at time of transplant (in absence of other apparent causes)	Not specified

CNI, Calcineurin inhibitor; uPCR, urine protein-to-creatinine ratio.

patients¹²⁴) and thymoma. It is most frequently observed in the nodular sclerosis subtype of Hodgkin lymphoma (71% of MCD cases secondary to this disease), typically coinciding with either the diagnosis or relapse of malignancy. In 38% of cases, glomerular disease precedes the diagnosis of lymphoma. Approximately half of MCD cases exhibit

steroid resistance but often remit with effective lymphoma treatment.¹²⁵

MCD has also been reported as a late complication following allogeneic hematopoietic stem cell transplantation (allo-HSCT), accounting for about 15% of glomerular diseases in this setting. MCD in these cases is usually seen

in patients with prior chronic graft-versus-host disease and typically presents as abrupt-onset NS, responding favorably to immunosuppressive therapy.^{126,127}

Drug-induced MCD is often linked to nonsteroidal anti-inflammatory drugs (especially fenoprofen, which has been withdrawn from the market), interferon, penicillins, lithium, and rifampicin. These cases may exhibit overlapping features of MCD and acute interstitial nephritis on biopsy, presenting clinically as NS accompanied by pyuria and renal insufficiency. Discontinuing the drug can lead to resolution of proteinuria; however, recovery of kidney function may take weeks to months and may be incomplete.^{28,129}

DIAGNOSTIC APPROACH

The diagnostic approach differs between children and adults.

Considering that the vast majority of childhood-onset NS is caused by INS, empiric treatment with high-dose steroids is recommended for the initial episode of NS. A rapid and CR, seen in >90% of pediatric cases, confirms the diagnosis of SSNS. Therefore kidney biopsy is not required in most children. However, it is warranted if an etiology other than MCD is suspected. Indications for biopsy include clinical features suggestive of a secondary cause (such as hematuria, hypertension, extrarenal manifestations, or low C3 levels) or AKI not attributable to hypovolemia. Additionally, a kidney biopsy is advised before initiating therapy with CNIs and following their prolonged use to assess histologic signs of nephrotoxicity. It is recommended also for patients with SRNS, except in syndromic or familial disease where genetic screening has higher diagnostic and prognostic role. Also, in patients with congenital nephrotic syndrome (younger than 3 months of age) and older infants (6–12 months) exhibiting steroid resistance, a genetic cause is likely in approximately 70% to 80% of cases, with next-generation sequencing being the preferred screening method. In pediatric populations, age thresholds do not significantly influence the decision to perform a kidney biopsy at disease onset.

In adults, diagnosis is based on kidney biopsy findings rather than empirical treatment response. Therapeutic decisions in adult-onset nephrotic syndrome require histologic diagnosis along with a comprehensive workup to rule out secondary causes.

TREATMENT

High-dose glucocorticoids remain the first-line treatment for both biopsy-proven MCD and pediatric INS. In cases of frequent relapses, steroid dependence, as well as infrequent relapses with significant glucocorticoid-related adverse effects, or relative contraindications to glucocorticoids, steroid-sparing agents are recommended. These include CNIs (such as cyclosporine or tacrolimus), cyclophosphamide, rituximab, mycophenolic acid analogs (MMF or MPS), and, in children, levamisole (Fig. 30.7–Pediatric Algorithm, Fig. 30.8–Adult Algorithm).

Although few randomized controlled trials directly compare these therapies, clinical practice guidelines recommend tailoring treatment based on the frequency and severity of relapses, glucocorticoid tolerance, prior therapy

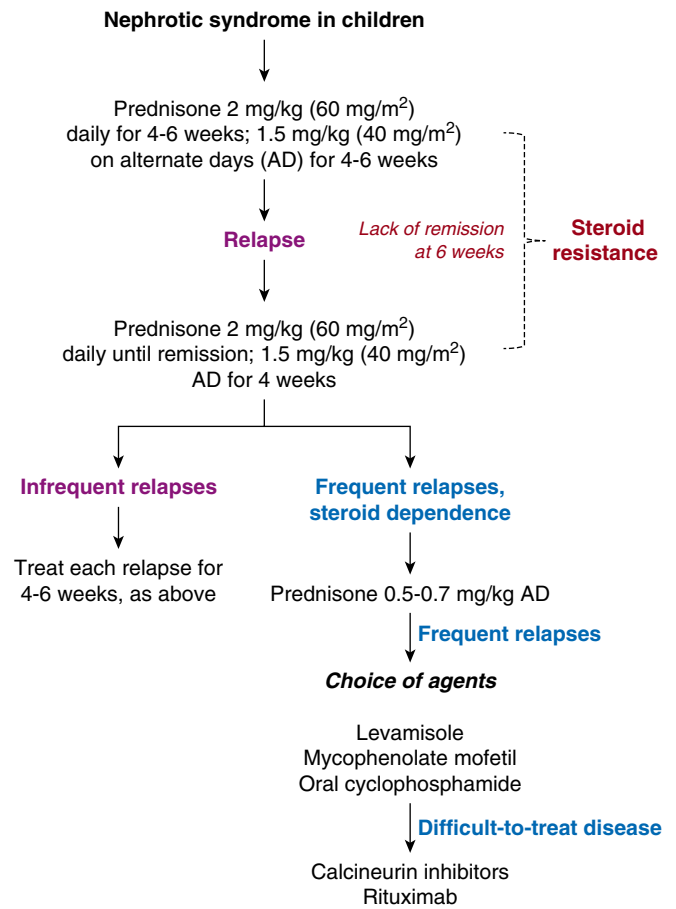


Fig. 30.7 Immunosuppressive treatment in nephrotic syndrome in children.

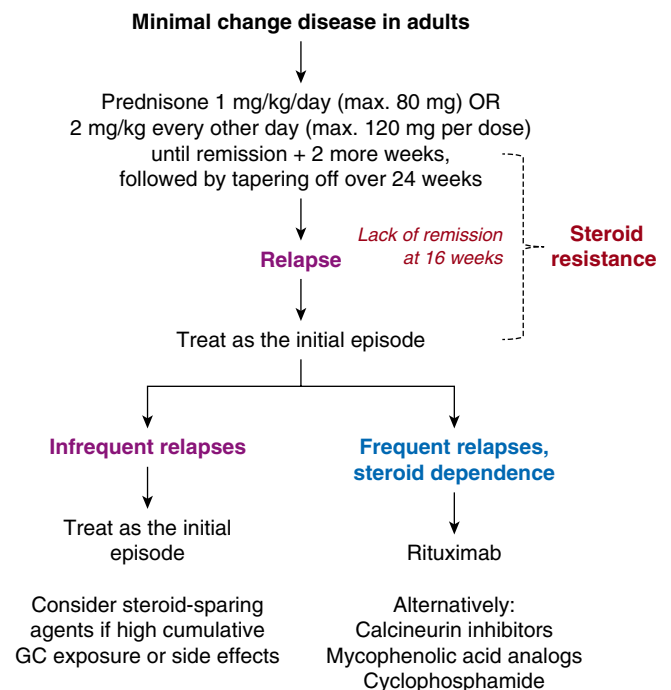


Fig. 30.8 Immunosuppressive treatment in minimal change disease in adults.

outcomes, safety profile, patient age, and patient and parental preferences (in pediatric cases), drug availability, and cost considerations.^{37,38,40,130} Table 30.2 provides a detailed overview of dosing, treatment duration, adverse effects, and necessary precautions for these agents in both children and adults.

TREATMENT OF IDIOPATHIC NEPHROTIC SYNDROME IN CHILDREN

Initial therapy consists of high-dose glucocorticoids, either based on body weight or on body surface area (see

Fig. 30.7)—both associated with similar outcomes.¹³¹ Therapy should be given for 8 to 12 weeks total at 60 mg/m²/day (or 2 mg/kg/day; maximum 60 mg) until remission (protein trace/nil for 3 consecutive days) and continued for 4 to 6 weeks, followed by 40 mg/m² (1.5 mg/kg, maximum 40 mg) on alternate days for 4 to 6 weeks. It has been shown that prolonged therapy, although it may delay the first relapse, does not affect the subsequent disease course (risk of subsequent or frequent relapses).^{38,40}

Relapses, occurring in >70% of children, mostly within year from diagnosis, are precipitated by minor, usually

Table 30.2 Immunosuppressive Agents for Frequently Relapsing and Steroid-Dependent Nephrotic Syndrome in Children and Minimal Change Disease/FSGS in Adults

Drug	Dose and Duration in Children	Dose and Duration in Adults	Adverse Effects; Precautions
Prednisone	0.5-0.7 mg/kg on alternate days for 6-12 months; the same dose may be administered daily for 5-7 days during infections	Not recommended	Weight gain, Cushingoid features; reduced height velocity; hypertension; raised intraocular pressure; cataract; impaired glucose tolerance Monitor anthropometry, blood pressure, glucose q 3-6 months; eye evaluation q 12 months
Levamisole	2-2.5 mg/kg on alternate days for 2-3 years	Not recommended	Neutropenia, vasculitic rash; raised transaminases, seizures Monitor blood counts q 2-3 months; ANCA, transaminases q 6 months
Cyclophosphamide	2-2.5 mg/kg/day orally for 8-12 weeks; or 500-750 mg/m ² IV every month for 6 months	2-2.5 mg/kg/day orally for 8 weeks	Leukopenia, alopecia, pigmentation of nails; risk of gonadal toxicity and malignancies Initiate therapy during remission; maintain hydration; monitor blood counts q 2 weeks; withhold drug during serious infection, or if leukocytes <4000/mm ³
Mycophenolate mofetil	800-1200 mg/m ² /day in divided doses for 2-3 years	2000 mg/day in divided doses	Abdominal pain, diarrhea; viral warts; leukopenia; elevated transaminases Monitor blood counts q 2-3 months; transaminases q 3-6 months
Calcineurin inhibitors Cyclosporine A	Administered for 2-3 years 4-5 mg/kg/day in divided doses; target 12-h trough 80-150 ng/mL	Administered for 1-2 years 3-5 mg/kg/day in divided doses, target initial 12-h trough level 150-200 ng/mL, dose reduction after withdrawal of glucocorticoids to <3 mg/kg/day	Both nephrotoxicity, hyperkalemia, hepatotoxicity Monitor creatinine, potassium at 2 weeks, then q 3-6 months; liver functions, glucose, uric acid, magnesium, and lipids
Tacrolimus	0.1-0.2 mg/kg/day in divided doses; target 12-h trough 5-7 ng/mL	0.05-0.1 mg/kg/day in divided doses, target initial 12-h trough level 4-7 ng/mL, dose reduction after withdrawal of glucocorticoids to <0.05 mg/kg/day	Specific to cyclosporine: gum hyperplasia, hirsutism; hypertension; dyslipidemia Specific to tacrolimus: Tremors, seizures, headache; diarrhea; glucose intolerance; hypomagnesemia
Rituximab	375 mg/m ² IV infusion, 2 doses 1 week apart (consider 1-2 additional doses if B cells not depleted)	For induction: 4x 375 mg/m ² weekly OR 1x 375 mg/m ² with repeating the dose after 1 week if CD19 >5 cells/mm ³ OR 2x 1.0 g, 2 weeks apart For relapses after RTX induction: 1x 375 mg/m ² OR 1x 1.0 g	Infusion reactions; serum sickness; bronchospasm; neutropenia; <i>P. jirovecii</i> pneumonia; acute lung injury; hypogammaglobulinemia; reactivation of hepatitis B, JC virus Rule out acute and chronic infections before therapy Monitor CD19 counts; blood counts; IgG levels; consider cotrimoxazole prophylaxis

Data from the guidelines of IPNA, ISPN, and KDIGO.³⁷⁻⁴⁰

upper respiratory tract, infections in almost half of cases. Proteinuria during infections may be transient, hence it is advisable to await resolution of infection, avoiding the need for corticosteroid therapy. However, 3+/4+ proteinuria that lasts for 3 days or longer is unlikely to resolve and is treated with prednisone at 60 mg/m²/day (or 2 mg/kg/day; maximum 60 mg) until remission (protein trace/nil for 3 consecutive days), followed by 40 mg/m² (1.5 mg/kg, maximum 40 mg) on alternate days for 4 weeks (see Fig. 30.7). Given the natural history of disease, relapses may continue to occur even while on immunosuppressive therapy; hence both sustained remission and infrequent relapses are considered acceptable and imply “stable remission.”

As mentioned earlier, almost half of pediatric patients with relapses show steroid dependency or frequently relapsing disease.^{37,38} Prolonged therapy with high-dose prednisone is associated with significant steroid toxicity, as well as behavior problems, cataracts, glaucoma, hypertension, avascular hip necrosis, and diabetes. Relapses are associated with significant complications including infections, thrombosis, and dyslipidemia. Therefore children with frequent relapses or steroid dependence require expertise in management, with an aim to reduce the frequency of relapses, without impairing normal growth and scholastic and extracurricular activities, and avoiding adverse effects of immunosuppressive medications. In particular, the subgroup of patients that continues to show frequent relapses (or infrequent relapses in presence of significant steroid toxicity) despite use of two or more steroid-sparing agents (such as levamisole, cyclophosphamide, or mycophenolate mofetil), termed “difficult-to-treat” steroid-sensitive disease, might merit consideration for use of more potent medications (see Fig. 30.7).³⁷ Table 30.2 summarizes recommendations on dose, duration and adverse effects, and precautions during therapy.

TREATMENT OF MINIMAL CHANGE DISEASE IN ADULTS

MCD treatment is recommended in all cases due to the low likelihood of spontaneous remission (unlike MN) and the significant morbidity associated with untreated nephrotic syndrome, such as infections and thromboembolic complications, among others. Treatment protocols for adults largely mirror those for children, with much of the evidence extrapolated from pediatric studies.

First-line therapy, as in pediatric INS, involves high-dose glucocorticoids (1 mg/kg/day, max. 80 mg/dose; or 2 mg/kg every other day, max. 120 mg/dose), which lead to CR within 16 weeks in more than 80% of adults (called steroid-sensitive MCD) (see Fig. 30.8). Glucocorticoid tapering is recommended starting 2 weeks after CR and continuing over 24 weeks. For individuals with relative contraindications to glucocorticoids, steroid-sparing drugs should be implemented, including CNIs, MPAA, and cyclophosphamide.⁴⁰ Emerging evidence from small studies also supports rituximab monotherapy as an effective option.¹³²⁻¹³⁴

Relapses are managed similarly to the initial episode, except for patients with frequent relapses or steroid dependence, accounting for approximately one-third of adult MCD cases. In these cases, steroid-sparing immunosuppressive agents are recommended to minimize

glucocorticoid exposure and reduce the risk of side effects (see Fig. 30.8). These agents are typically initiated after achieving CR with high-dose glucocorticoids, and the glucocorticoids are tapered to discontinuation over 2 to 4 weeks.⁴⁰ The Immunonephrology Working Group of the European Renal Association (IWG-ERA) recommends rituximab as the preferred option due to its long-lasting effect, convenient biannual administration, and favorable safety profile without metabolic side effects.¹³⁵ However, given the comparable efficacy of available treatments, the choice of therapy should consider patient preferences, drug side effect profiles, availability, and cost.

OUTCOMES

Before the advent of glucocorticoid therapy, mortality in children with SSNS exceeded 30%, largely due to disease-related complications.³² The introduction of immunosuppressive treatment has reduced this number to nearly zero; however, the recurrent nature of the disease and the adverse effects of therapy remain significant clinical challenges.

Steroid toxicity is a major concern, particularly in patients with frequent relapses or steroid dependence. Complications include short stature in approximately 8.2% of those with childhood-onset disease,¹³⁶ overweight or obesity in >40%,¹³⁷ hypertension in one-third,⁶⁹ metabolic bone disease in up to 40%,¹³⁸⁻¹⁴¹ diabetes mellitus in 12%,³¹ and ocular complications in about 20%,¹⁴² among others.

RENAL OUTCOMES

Renal outcomes in MCD and INS are excellent, with <5% of patients progressing to ESKF. The incidence of ESKF is estimated at 1.34 to 2.22 per 100 patient-years.¹²² Long-term prognosis depends on the response to immunosuppressive therapy and worsens in patients with secondary steroid resistance or histologic progression to FSGS, where the 10-year risk of ESKF can reach 70%. Outcomes in steroid-resistant nephrotic syndrome in children and FSGS in adults are discussed in detail later.

FOCAL SEGMENTAL GLOMERULOSCLEROSIS

FSGS is a histologic pattern of injury characterized by sclerotic lesions involving <50% of glomeruli (focal lesions) and affecting only a segment of the glomerular tuft (segmental lesions) on light microscopy, as well as negative staining for immunoglobulins and complement on immunofluorescence, along with variable degrees of podocyte FPE observed on electron microscopy.

It is one of the most common histologic patterns of glomerular injury and a leading cause of ESKF among glomerular diseases, second only to immunoglobulin A (IgA) nephropathy. Importantly, FSGS represents a nonspecific lesion with highly heterogeneous etiology including primary, genetic, and secondary forms. Determining the underlying cause is challenging; diagnosis requires a comprehensive assessment integrating clinical presentation, histopathologic findings, and thorough evaluation for secondary and genetic causes. Nonetheless, a considerable

proportion of cases remain of undetermined origin. These complexities make FSGS one of the most diagnostically and therapeutically challenging glomerular diseases.

EPIDEMIOLOGY

FSGS is the most frequent pattern of injury worldwide, except in Asia, where IgA nephropathy is more prevalent. FSGS accounts for 10% to 20% of all biopsy diagnoses and 20% to 40% in adults.^{79,143} Among patients with nephrotic syndrome, FSGS is diagnosed in 35% of cases overall, rising to approximately 80% within the African American population.¹⁴⁴ In children, the FSGS pattern is less commonly detected, except in SRNS patients, where it is observed in 40% to 50% of cases.¹⁴⁵

The annual incidence of FSGS varies significantly, ranging from 0.8 to 22.9 cases per 100,000 individuals.^{80,143,146} However, a consistent upward trend has been observed, attributed to increases in both primary and secondary forms of FSGS. The rising prevalence of secondary FSGS is linked to the obesity epidemic, which has driven a more than 10-fold increase in secondary adaptive FSGS cases over the past 3 decades.^{143,146-148} In contrast, the increase in primary FSGS is thought to result from the wider availability of kidney biopsy procedures and possibly greater exposure to environmental factors contributing to podocyte injury.^{79,149} The prevalence of FSGS varies geographically, with individuals of African descent being disproportionately affected, exhibiting a fivefold higher frequency compared with the Caucasian population.¹⁴⁶ This disparity is associated with the higher prevalence of high-risk apolipoprotein L1 (*APOL1*) alleles in populations of African ancestry, which increase the risk of secondary FSGS by 17-fold.¹⁵⁰

ETIOLOGY AND PATHOPHYSIOLOGY

FSGS is a nonspecific lesion resulting from podocyte injury or dysfunction. Primary, immune-mediated disease targeting podocyte antigens accounts for only 40% to 45% of cases,²¹ with the remaining cases primarily caused by maladaptive responses to mechanical stress (adaptive FSGS), inherited defects in genes encoding key podocyte proteins (genetic FSGS),¹⁵¹ viral infections (e.g., human immunodeficiency virus [HIV], acute respiratory syndrome coronavirus 2 [SARS-CoV-2]), and drug toxicities (e.g., heroin, interferon, and bisphosphonates) (Table 30.3).

PRIMARY FOCAL SEGMENTAL GLOMERULOSCLEROSIS

Primary FSGS is estimated to account for 40% to 45% of all FSGS cases. Growing evidence supports the hypothesis that this immune-mediated podocytopathy shares a similar pathogenesis with MCD/INS and is considered a histologic progression, with a slower and poorer response to treatment compared with MCD, as well as a higher risk of progression to ESKF.¹⁵² However, the data regarding the prevalence of antinephrin antibodies are conflicting. Antibodies have been detected in only 9% of presumed primary FSGS cases in the landmark study,¹⁰⁶ whereas a study of patients with post-transplant recurrence of presumed primary FSGS has found serum antinephrin antibodies and IgG deposits colocalizing with nephrin on graft biopsy in all cases.¹⁰⁸ These findings suggest that further investigation is warranted.

GENETIC FOCAL SEGMENTAL GLOMERULOSCLEROSIS

Among genetic forms of FSGS, both inherited and sporadic cases can be distinguished, with presentations either restricted to the kidneys or syndromic (see also Chapter 44). Additionally, polymorphisms such as those in the *APOL1* gene have been identified as contributors to the development of secondary FSGS (see later).

A hallmark of genetic nephrotic syndrome is its lack of response to immunosuppression (SRNS course). Genetic causes account for about 15% of SRNS in children,¹²³ with mutations in podocyte-associated genes present in nearly 50% of patients with a family history of SRNS and in 10% to 30% of nonfamilial cases.^{153,154} The likelihood of identifying a monogenic cause of SRNS decreases inversely with age.¹⁵³⁻¹⁵⁵ While two-thirds of those presenting before 1 year of age have monogenic disease, the chance of identifying a genetic cause for SRNS decreases substantially to 25% in children aged 1 to 6 years, 18% in those aged 7 to 12 years, 11% in adolescents aged 13 to 18 years,¹⁵⁵ and 8% to 14% in adult-onset FSGS.¹⁵⁶

Rapid advances in next-generation sequencing have facilitated the identification of numerous novel causative genes. More than 80 genes are associated with monogenic SRNS, chiefly encoding structural elements of the podocyte slit-diaphragm or cytoskeleton (including *NPHS1*, *NPHS2*, *CD2AP*, *TRCP6*, and *ACTN4*); GBM proteins (*LAMB2*, *COL4A5*); mitochondrial genes (*COQ2*); or nuclear transcription factors (*WT1*, *LMX1B*) (Table 30.4).^{39,151,157} These genetic variations may be acquired in an autosomal recessive (*NPHS1*, *NPHS2*, *PLCE1*) or dominant (*INF2*, *TRCP6*, *ACTN4*) manner, or both (*WT1*). In addition, several monogenic mutations result in syndromic forms of SRNS (Table 30.5). Reports have highlighted a high frequency of mutations in genes encoding type IV collagen (*COL4A3*, *COL4A4*, *COL4A5*), often as phenocopies of SRNS,¹⁵⁸⁻¹⁶⁰ with or without coexisting laminin alpha 5 (*LAMA5*) variants that act as potential effect modifiers.¹⁶¹ Of note, although the FSGS pattern predominates among genetic forms, MCD and diffuse mesangial sclerosis (DMS) are also seen. DMS is a histopathology observed in young children, characterized by progression to ESKF in early childhood. More than 80% of children with DMS have identifiable causal variants in *NPHS2*, *WT1*, *PLCE1*, or *LAMB2*.¹⁶²

SECONDARY FOCAL SEGMENTAL GLOMERULOSCLEROSIS

Adaptive FSGS arises from glomerular maladaptation to reduced nephron numbers (e.g., due to low birth weight and solitary kidney) or excessive stress on a normal nephron population (e.g., obesity, high-protein diet, reflux nephropathy, and advanced stages of CKD of all causes) (see Table 30.3). These conditions disrupt the balance between glomerular load and capacity, leading to glomerular hypertension and hyperfiltration. The resulting compensatory glomerular hypertrophy increases the filtration surface area to counteract hypertension. However, this adaptation forces podocytes to stretch to cover the expanded surface, weakening their attachment to the GBM, ultimately causing detachment and loss. Hyperfiltration further exerts mechanical strain on podocytes, especially in the perihilar segments of the glomerular tuft, where blood flow is most disturbed. This mechanical stress initiates a loss

Table 30.3 Etiologies of Focal Segmental Glomerulosclerosis (FSGS) Pattern of Injury

Etiology	Example	Clinical Presentation	Histology
Primary			
Immune-mediated podocytopathy	Nephrin-associated FSGS	Steroid-sensitive nephrotic syndrome	LM: tip lesion, cellular variant EM: diffuse FPE (≥80%)
Genetic			
Renal limited (familial or sporadic)	See Table 30.5	Steroid-resistant nephrotic syndrome	LM: variable EM: variable
Syndromic	See Table 30.6		
Secondary			
Adaptive	Reduced nephron mass (low birth weight, renal agenesis/dysplasia, oligomeganephronia) Loss of nephron mass (nephrectomy) Increased stress in normal nephron count (obesity, high-protein diet, reflux nephropathy, sickle cell anemia, any advanced kidney disease, cyanotic congenital heart disease, hypoxic pulmonary disease)	Subnephrotic or neprotic-range proteinuria without NS, slowly progressing chronic kidney disease	LM: perihilar variant or mixed, glomerulomegaly, FSGS lesions in a few glomeruli only (average of 10%) EM: mild, segmental FPE (<80%; e.g., in 40% in obesity-related FSGS)
Viral infections	HIV parvovirus B19, SARS-CoV2 CMV, EBV, HCV	Nephrotic syndrome, acute kidney injury with rapidly progressive decline in kidney function, high risk of ESKF	LM: collapsing FSGS variant, tubulointerstitial inflammation, microcystic tubular changes EM: diffuse FPE
Drug toxicity	Nonsteroidal anti-inflammatory drugs (NSAIDs) Interferon Penicillin Pamidronate Rifampicin mTOR inhibitors Calcineurin inhibitors Anthracyclines Anabolic steroids Heroin	Variable clinical presentation, including severe nephrotic syndrome, AKI	LM: variable, including collapsing variant with concomitant acute interstitial nephritis EM: variable
APOL1-related FSGS		Variable clinical presentation	LM: collapsing FSGS variant EM: variable
FSGS of Undetermined Cause			
No identifiable cause	Unknown	Variable clinical presentation	LM: variable EM: variable

of interdigitating foot processes appearance and ultimately leads to FPE, particularly in the perihilar region.^{156,163} FPE occurs progressively and heterogeneously, involving only a part of glomerular surface (segmental FPE) (e.g., in 25% of reflux nephropathy-associated FSGS and 40% in obesity-associated FSGS).^{148,156}

Additionally, individuals of African ancestry may carry sequence variants in the *APOL1* gene, G1 and G2, that confer the risk of FSGS in a non-Mendelian fashion. Apolipoprotein L1, the gene product, is a protease that protects against *Trypanosoma brucei*, the parasite responsible for African sleeping sickness. These high-risk variants are found in approximately 35% of the African American population and 25% to 50% of the African population.⁷⁹ Inheriting two such

variants raises the risk for three distinct APOL1-mediated kidney diseases: hypertension-associated end ESKF with global glomerular sclerosis (a 7- to 10-fold increase), FSGS (around 17-fold), and HIV-associated nephropathy (29- to 89-fold) often manifesting as collapsing FSGS.¹⁵⁰ ApoL1 disease is discussed in more detail in [Chapters 44 and 75](#).

Among viral infections, the pathogenesis of HIV-associated FSGS is the most extensively studied. HIV-associated nephropathy (HIVAN) predominantly affects individuals with high-risk *APOL1* alleles, accounting for its disproportionately high prevalence among people of African and African American ancestry.^{164,165} The pathogenesis involves direct podocyte infection by the virus, resulting in actin cytoskeleton disruption and FPE. This damage is

Table 30.4 Genes Associated With Steroid Resistant Nephrotic Syndrome Grouped According to Location of Corresponding Proteins in Podocyte

Location of Proteins	Genes
Slit-diaphragm associated proteins	<i>NPHS1, NPHS2, PLCE1, CD2AP, TRPC6, CRB2, FAT1</i>
Cytoskeletal and membrane proteins	<i>ACTN4, INF2, MYH9, MYO1E, MAGI2, ANLN, ARHGAP24, ARHGDI, KANK, 1/2/4, SYNPO, PTPRO, EMP2, APOL1, CUBN, PODXL, TNS2, DLC1, CDK20, ITSN1, ITSN2</i>
Basement membrane-associated proteins	<i>LAMB2, ITGB4, ITGA3, COL4A3/4/5, GPC5, CD151</i>
Mitochondrial proteins	<i>COQ2, COQ6, PDSS2, ADCK4, MTTL1</i>
Lysosomal and endocytic proteins	<i>SCARB2, OCRL1</i>
Metabolic and cytosolic proteins	<i>ZMPSTE24, PMM2, ALG1, TTC21B, CFH, DGKE</i>
Nuclear pore proteins and transcription factors	<i>NUP93, NUP205, XPO5, WT1, LMX1B, SMARCL1, E2F3, NX5, PAX2, WDR73</i>

further aggravated by cytokine release, including VEGF, which exacerbates podocyte injury.¹⁶⁵ Histologic changes predominantly manifest as the collapsing variant of FSGS, accompanied clinically by nephrotic syndrome and severe AKI.

SARS-CoV-2-associated FSGS, termed COVID-associated nephropathy (COVAN), also presents histologically as collapsing FSGS and mirrors the pathogenesis of HIV-associated collapsing FSGS. This includes a combination of direct podocyte infection and cytokine-mediated injury, particularly among individuals with high-risk *APOL1* gene polymorphisms. Other viral infections, such as cytomegalovirus, Epstein-Barr virus, parvovirus B19, and hepatitis C virus, have also been linked to FSGS. However, the pathogenesis of FSGS associated with these infections remains poorly understood. Infections are discussed in more detail in [Chapters 34 and 59](#).

The mechanisms underlying drug-associated FSGS remain controversial. This pattern of injury has been reported in association with several agents, including bisphosphonates (collapsing and other variants, as well as MCD lesions), interferon, lithium (also linked to MCD), and heroin abuse, among others.⁷⁹

PATHOLOGY

FSGS is defined by focal and segmental glomerular sclerosis or consolidation observed on light microscopy. Histologic changes are focal and, in the early stages, predominantly affect juxtamedullary glomeruli, which may be missed in biopsy that are either too small or taken from superficial areas. A diagnosis requires an evaluation of consecutive

sections selected from 12 to 15 serial sections, each containing 8 to 9 glomeruli,¹⁶⁶ while optimal samples should include at least 20 glomeruli.¹⁶⁷

The distribution of sclerotic lesions in FSGS varies. According to the Columbia classification, five histologic variants are defined on the basis of the predominant location of changes: tip lesion, collapsing, perihilar, cellular, and not otherwise specified (NOS).¹⁴⁵ The clinical utility of this classification is, however, limited, as the NOS variant is the most common, none of the variants are pathognomonic for distinguishing primary from secondary FSGS, and repeated kidney biopsies reveal variant changes in approximately half of cases, mostly evolution to NOS type.¹⁶⁸

- The tip lesion variant, accounting for approximately 17% of cases, is characterized by sclerosis of the glomerular segment adjacent to the origin of the proximal tubule, opposite the hilum. It is often associated with primary FSGS and has the most favorable renal prognosis. The collapsing variant, observed in 11% of cases, involves segmental collapse of glomerular capillaries with obliteration of capillary lumens, forming pseudocrescents. This variant is associated with significant tubulointerstitial damage, including tubular microcystic dilatation and interstitial inflammation. It represents the most aggressive histologic form, often progressing rapidly to irreversible injury: glomerulosclerosis, tubular atrophy, and interstitial fibrosis. Collapsing FSGS is most frequently linked to secondary causes, such as HIV or SARS-CoV2 (coronavirus disease 2019, COVID-19) infection (see [Chapter 59](#)), and drug toxicity, although it can also occur in primary, immune-mediated FSGS.
- The perihilar variant is seen in 26% of cases and is characterized by lesions predominantly located in the perihilar region, often accompanied by hyalinosis. This variant is typically associated with an adaptive response to hyperfiltration or reduced nephron mass. The perihilar distribution of FSGS lesions corresponds to the region of highest ultrafiltration pressure¹⁴⁸ and is often associated with glomerulomegaly (average glomerular diameter of 226 μ m compared with 169 μ m in matched healthy controls), podocyte hypertrophy, and detachment. These changes typically affect a small proportion of glomeruli (approximately 10%).¹⁴⁷ On electron microscopy, FPE is usually segmental, involving only 25% to 40% of the glomerular surface.^{147,148,169} Although generally linked to adaptive FSGS, perihilar changes can predominate in certain monogenic forms of childhood nephrotic syndrome.¹⁷⁰ The cellular variant, which is rare in adults (3%) but more frequently observed in pediatric SRNS cases (21%),¹⁷⁰ features proliferative lesions that affect multiple regions of the glomerular tuft rather than being confined to the tip.
- The NOS category accounts for 42% of cases and is used for lesions that lack the distinctive features of other variants.

In all histologic variants of FSGS, nonsclerotic glomeruli and segments typically show no staining for immunoglobulins

Table 30.5 Genes Associated With Syndromic Forms of Steroid Resistance and Key Phenotypic Features

Syndrome	Gene	Protein	Function	Inheritance	Associated Features; Histology
Denys-Drash syndrome Frasier syndrome	<i>WT1</i>	Wilms tumor protein	Podocyte development	AD	Male pseudohermaphroditism, Wilms tumor; DMS Male pseudohermaphroditism, gonadoblastoma; FSGS
Pierson syndrome	<i>LAMB2</i>	Laminin β -2	Links GBM to podocyte cytoskeleton	AR	Microcoria, abnormal lens; DMS
Schimke immune-osseous dysplasia	<i>SMARCA1</i>	SWI/SNF-related, matrix-associated, actin-dependent regulator of chromatin, subfamily A-like protein 1	Chromatin bundling and gene transcription	AR	Bone dysplasia, immune deficiency, ischemic cerebral lesions; FSGS
Nail-patella syndrome	<i>LMX1B</i>	LIM homeodomain transcription factor 1, β	Podocyte and GBM development	AD	Hypoplastic nails and patellae, iliac horns; FSGS
Charcot-Marie-Tooth disease	<i>INF2</i>	Inverted formin 2	Actin regulation	AD	Neuropathy, deafness; FSGS
Galloway-Mowat syndrome	<i>WDR73</i> , <i>WDR7</i> , <i>OSGEP</i> , <i>TP53RK</i> , <i>TPRKB</i> , <i>LAGE3</i>	WD40 repeat-containing protein	Unknown	AR	Microcephaly, intellectual disability, hiatal hernia, optic atrophy; FSGS
Mitochondrial respiratory-chain disease	<i>MTTL1</i>	Mitochondrial tRNA leucine 1	Mitochondrial tRNA	Maternal	MELAS (mitochondrial encephalomyopathy, lactic acidosis, and stroke-like symptoms), diabetes mellitus, deafness; FSGS
	<i>COQ2</i>	Polyprenyltransferase	Coenzyme Q10 biosynthesis	AR	Encephalomyopathy, hypotonia, seizures, lactate acidosis; FSGS
	<i>COQ6</i>	Ubiquinone biosynthesis monooxygenase COQ6	Coenzyme Q10 biosynthesis	AR	Sensorineural deafness; FSGS
	<i>PDSS2</i>	Decaprenyl diphosphate synthase, subunit 2	Coenzyme Q10 biosynthesis	AR	Encephalomyopathy, hypotonia, seizures, lactate acidosis; FSGS

AD, Autosomal dominant; AR, autosomal recessive; DMS, diffuse mesangial sclerosis; FSGS, focal segmental glomerulosclerosis; GBM, glomerular basement membrane.

or complement on immunofluorescence microscopy. Sclerotic segments, however, may exhibit irregular staining for C3 and IgM, reflecting nonspecific protein trapping.

The primary finding on electron microscopy in FSGS is podocyte FPE, the extent of which provides insight into the underlying pathogenesis. Diffuse and widespread FPE, defined as affecting $\geq 80\%$ of the glomerular surface, is indicative of immune-mediated (primary) FSGS, whereas mild or segmental FPE is more suggestive of secondary causes.^{79,169} Notably, exceptions to this pattern occur in HIV-associated, drug-induced, and certain genetic forms of FSGS, where diffuse FPE may be observed despite a nonprimary etiology. Furthermore, biopsies performed during immunosuppressive therapy for immune-mediated FSGS may display segmental FPE, reflecting partial recovery and complicating interpretation. Of note, accurate

assessment of FPE requires evaluating intact, nonsclerotic glomeruli, as sclerotic glomeruli can misleadingly indicate widespread FPE in secondary forms of FSGS.¹⁶³

CLINICAL PRESENTATION AND DISEASE COURSE

The clinical presentation of FSGS varies widely due to its heterogeneous etiology. The hallmark feature is proteinuria, which can range from subnephrotic levels to nephrotic-range proteinuria and severe nephrotic syndrome. Hematuria is observed in more than 50% of patients, while hypertension is present in approximately one-third of cases, particularly in adults. About 33% of patients present with some degree of kidney function impairment and the progression to ESKF is influenced by the underlying etiology and the effectiveness of treatment in reducing proteinuria.^{144,171}

Immune-mediated (primary) FSGS shares a clinical presentation similar to MCD/INS, characterized by the abrupt onset of severe nephrotic syndrome. Nephrotic syndrome is observed in all cases with secondary causes excluded and diffuse FPE ($\geq 80\%$ of the glomerular surface)¹⁴³ but only in 50% when the FPE criterion is not applied. Immunosuppressive therapy induces complete or partial remission in 60% to 80% of cases.¹⁷²⁻¹⁷⁴ Relapses are less frequent than in MCD/INS, occurring in 30% to 60% of cases.^{175,176} Achieving remission, even partial, is critical for long-term kidney outcomes, underscoring the importance of accurately identifying immune-mediated FSGS and implementing effective treatment.

In children, monogenic forms typically present as NS and are phenotypically difficult to distinguish from INS/MCD at presentation, but they show initial steroid resistance and do not respond to immunosuppressive medications. SRNS caused by mutations in autosomal recessive genes tends to present early in childhood and manifests as severe nephrotic syndrome and progressive kidney disease. The most common is a mutation in gene-encoding podocin (*NPHS2*), followed by nephrin (*NPHS1*, also termed a congenital nephrotic syndrome of the Finnish type). In contrast, late childhood- or adult-onset monogenic FSGS is more often due to autosomal dominant genes showing varying penetrance and severity,¹⁵¹ presenting with varying clinical course but often with milder proteinuria (<5 g/day). Syndromic genetic forms of SRNS in children and FSGS in adults may also present with extrarenal manifestations associated with the specific underlying genetic mutation (see Table 30.5).

In adaptive FSGS, proteinuria is generally subnephrotic or nephrotic range without hypoalbuminemia or other nephrotic syndrome features, aiding in differential diagnosis.^{169,177} Light microscopy may reveal glomerulomegaly, perihilar FSGS lesions, and segmental FPE. Renal insufficiency progresses slowly.¹⁴⁸ APOL1-mediated FSGS primarily affects individuals of African ancestry, often young, with collapsing variant on light microscopy, and an aggressive disease course characterized by an annual eGFR decline of 10% to 18%¹⁷⁸ and a high likelihood of progressing to ESKF, with a risk as high as 15%.¹⁵⁰ Similarly, an aggressive course is observed in HIVAN and COVAN, viral-associated FSGS, and also seen in individuals carrying high-risk *APOL1* alleles.

DIAGNOSTIC APPROACH

The diagnosis of FSGS integrates medical history with clinical and pathologic findings. A proposed diagnostic algorithm is outlined in Fig. 30.9.

Key considerations include adaptive causes (e.g., low birth weight, comorbidities and high body mass index); secondary causes (e.g., medications, substance abuse, and viral infections); and genetic causes (e.g., family history of SRNS or syndromic features like hearing loss in Alport syndrome or cardiomyopathy in Fabry disease) (see Table 30.5). Diagnostic guidance is additionally provided by clinical presentation (nephrotic syndrome, nephrotic-range proteinuria, or subnephrotic proteinuria); the dynamics of proteinuria progression; changes in renal function; degree of FPE on electron microscopy; and, in some cases, histologic

variants of FSGS. Future developments, such as diagnostic tests for serum antinephrin antibodies, may further streamline differential diagnosis.

In presumed primary immune-mediated FSGS, characterized by nephrotic syndrome, diffuse FPE, and absence of secondary causes, immunosuppressive therapy is recommended. A reduction in proteinuria supports continued therapy, while nonresponse necessitates treatment intensification.^{152,179} Persistent nephrotic syndrome beyond 16 weeks warrants genetic testing.

For patients with subnephrotic or nephrotic proteinuria without hypoalbuminemia and segmental FPE, adaptive FSGS is the most likely diagnosis and immunosuppressive therapy is not recommended.

In cases with inconclusive findings (e.g., NS with segmental FPE or subnephrotic proteinuria with diffuse FPE), genetic testing should be pursued. While the likelihood of identifying a causative mutation in adults is relatively low (8%–14%),¹⁵⁶ confirming a genetic etiology has significant implications including enabling genetic counseling, stratifying the risk for allograft recurrence, and facilitating donor screening (to identify asymptomatic carriers among relatives) in cases of progression to ESKF.^{40,157,180} Genetic testing should begin with next-generation sequencing panels tailored to the patient's age and ethnicity, with African ancestry panels including *APOL1* variants.^{40,156} If results are inconclusive, whole-exome sequencing (WES) or whole-genome sequencing (WGS) may be considered. The diagnosis and management of complex cases should be conducted at specialized centers with expertise in FSGS.

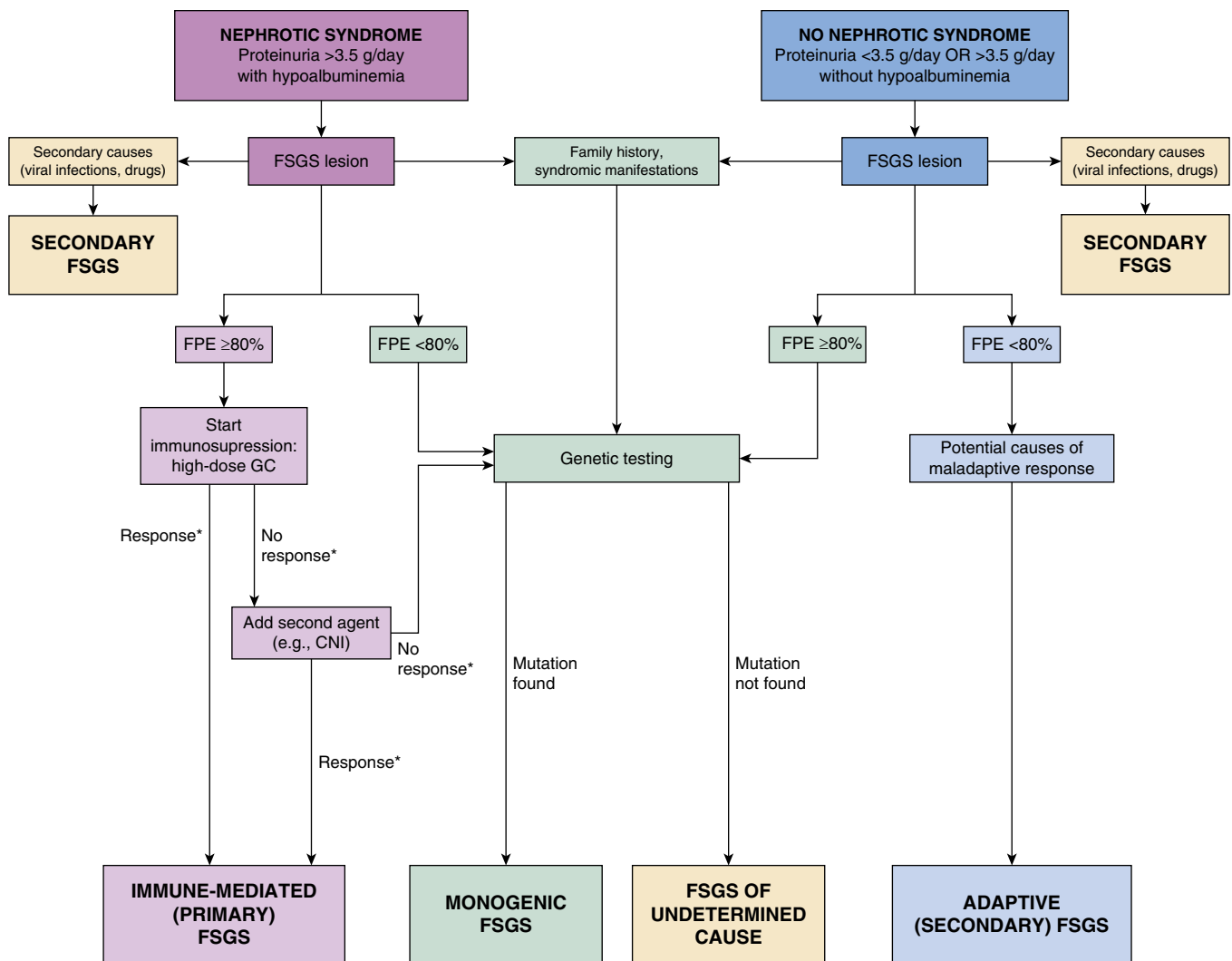
TREATMENT

Immunosuppressive therapy is indicated exclusively for immune-mediated (primary) FSGS and should not be offered to patients with non-immune-mediated forms, such as genetic, secondary FSGS, or FSGS of undetermined cause.

IMMUNOSUPPRESSIVE TREATMENT

First-line treatment for immune-mediated (primary) FSGS mirrors that for MCD, with high-dose glucocorticoids being recommended. However, FSGS patients generally show a slower and less frequent response to therapy, with remission rates of approximately 60% to 70%,⁷⁹ often only partial (observed in one-third to half of patients). Optimal management for patients with delayed or absent response to glucocorticoids remains unclear. According to KDIGO guidelines, high-dose glucocorticoids should be continued for up to 16 weeks, and in cases of failure to achieve at least partial remission, a CNI should be added.⁴⁰ Some studies suggest earlier initiation of additional immunosuppressive drug after 8 weeks in cases where proteinuria reduction is $<20\%$, allowing to limit glucocorticoid exposure and increase the response rate.¹⁵²

CNIs are also recommended as an alternative to high-dose glucocorticoids in patients with relative contraindications to steroids, typically in combination with low-dose glucocorticoids. The therapeutic effect of CNIs may take 4 to 6 months to manifest; hence CNI resistance should not be diagnosed before a minimum of



* Response is defined as reduction of proteinuria >20% in 8 weeks of GC treatment OR at least partial remission in 16 weeks of GC treatment

Fig. 30.9 Algorithm for differential diagnostic of focal segmental glomerulosclerosis lesion. CNI, Calcineurin inhibitor; FPE, foot process effacement; FSGS, focal segmental glomerulosclerosis; GC, glucocorticoid.

6 months of therapy. CNIs are effective in 50% to 70% of steroid-resistant cases,^{175,176,181,182} and treatment should be maintained for at least 12 months. Discontinuation of CNIs, similar to that seen in MCD, is associated with a significant relapse rate, reported in up to 30% to 60% of patients. Moreover, transient, reversible declines in eGFR are observed in 16% to 40% of patients receiving CNI therapy.^{176,182} Relapsing FSGS cases should be managed following MCD guidelines.⁴⁰

Patients with treatment-resistant FSGS should be referred to specialized centers and considered for enrollment in clinical trials. Current options for refractory cases include rituximab, MMF, and cyclophosphamide, though the use of the latter is limited by its unfavorable toxicity profile. Novel immunosuppressive agents under investigation for primary FSGS include obinutuzumab—a next-generation anti-CD20 antibody offering more potent and prolonged B-cell depletion (NCT04983888); adalimumab—an anti-TNF- α antibody (NCT04009668); SAR 442970—OX40 ligand

and TNF- α inhibitor; frexalimab—anti-CD40L antibody; and rilzabrutinib—Bryton tyrosine kinase inhibitor (NCT06500702); among others.^{183,184}

CONSERVATIVE MANAGEMENT

All FSGS patients including immune-mediated and non-immune-mediated forms (e.g., genetic, secondary, and undetermined) should receive comprehensive supportive care (as described above in the “Conservative Management” section of “Management of Patients with Nephrotic Syndrome” and summarized in Fig. 30.2) and targeted treatment of underlying causes where feasible (e.g., weight reduction in obesity-associated FSGS, urologic interventions in reflux nephropathy).

OUTCOMES

KIDNEY OUTCOMES

FSGS accounts for approximately 3% of all cases of ESKF.¹⁴⁶ Kidney survival is closely linked to the etiology, degree of

proteinuria, efficacy of therapy in reducing proteinuria, and kidney function at diagnosis.¹⁴⁴ Registry data show that proteinuria >1.5 g/g increases the risk of kidney failure or death by approximately 2.3-fold (HR 2.34, 95% CI: 1.99–2.74).²⁰ Additionally, a urine albumin-to-creatinine ratio (uACR) >0.7 g/g is associated with an unfavorable kidney outcome (defined as decline in eGFR >40%, eGFR <15 mL/min/1.73 m², or the initiation of kidney replacement therapy) by 5.27-fold in one study.²¹

Primary FSGS, particularly in cases of refractory NS, is considered a high-risk factor for ESKF showing a rate of progression to kidney failure of 24% in 5 years, and 43% in 10 years of follow-up.¹⁸⁵ In contrast, patients who achieve remission (either partial or complete) have a more favorable prognosis, with only 8% to 15% progressing to ESKF within 5 years and less than 8% to 25% within 10 years.^{144,171,185} Similar findings are reported in pediatric studies on SRNS, where FSGS is the most common histologic diagnosis (55%–60% of cases).¹²³ Among patients with undifferentiated or nongenetic SRNS, response to immunosuppression predicts kidney survival, with significantly higher risk of kidney failure in patients with persistent nonresponse.^{50,123,180,186,187} The 10-year ESKF rate was 6% for those achieving CR, 28% for those achieving partial remission, and 57% for nonresponders.¹²³

In contrast, secondary FSGS, typically presenting as subnephrotic or nephrotic-range proteinuria without the full nephrotic syndrome, generally follows a slower course, with less than 15% progressing to ESKF over 10 years.^{144,171} However, kidney failure eventually occurs in a significant proportion of patients including 10% to 33% of those with obesity-related FSGS.^{79,188} On the other hand, APOL1-mediated FSGS, presenting as the collapsing variant, is characterized by rapid decline in kidney function and a 4-fold higher risk of progression to ESKF compared with other forms of FSGS, with a risk as high as 15%¹⁵⁰ and peak occurrence of ESKF in individuals in their fifth decade.¹⁴⁶

Data on genetic forms of FSGS come mainly from pediatric studies on SRNS, showing a poor renal prognosis with a 2.9-fold higher odds of kidney failure compared with nongenetic cases.⁵⁰ The progression to ESKF is 73% within 10 years and 83% within 15 years in these patients.¹²³

POST-TRANSPLANT RECURRENCE OF FOCAL SEGMENTAL GLOMERULOSCLEROSIS

Immune-mediated FSGS is associated with a high recurrence rate after kidney transplantation, reaching 30% to 60%, and even 80% in cases of retransplantation due to graft loss resulting from FSGS recurrence.⁹² A similar frequency has been reported in children with SRNS. A systematic review and meta-analysis of 8 studies involving 581 children with SRNS estimated the overall risk of recurrence at 39% (95% CI: 34%–44%),¹⁸⁹ with the risk increasing to 61% (95% CI: 53%–69%) in patients without an identified causative genetic mutation, particularly in those with secondary SRNS, where the recurrence risk can be as high as 60% to 80%.^{190,191}

Proteinuria recurrence after renal transplantation in immune-mediated (primary) FSGS typically occurs within hours to weeks post transplant, with an average time to recurrence estimated at 1.5 months.⁹⁰ This recurrence is associated with an increased risk of delayed graft function

and both early and late graft loss, with rates reaching 40% to 50% over 5 years.^{90,92} Data from a small study of 11 patients indicate punctate IgG deposits colocalizing with nephrin and the presence of antinephrin antibodies in the sera of these patients.¹⁰⁸ Therapy usually comprises a combination of plasma exchange or immunoadsorption, high doses of CNIs, and corticosteroids, with or without one to two doses of rituximab.¹⁹² Additional strategies that have been used successfully include addition of ACEI or ARB, high-dose corticosteroid pulses, and LDL apheresis.¹⁹³ In contrast, maladaptive FSGS does not recur. Similarly, no recurrence is observed in grafts affected by FSGS associated with monogenic diseases.

MEMBRANOUS NEPHROPATHY

Membranous nephropathy (MN) is a glomerular disease characterized by diffuse, global GBM thickening, spikelike morphology on Jones silver stain, granular IgG and C3 deposits on immunofluorescence, and subepithelial electron-dense deposits with diffuse podocyte FPE on electron microscopy. It is the leading histologic diagnosis in nondiabetic adults with nephrotic syndrome,⁷³ accounting for 25% of cases,^{194,195} and ranks as the second or third most common glomerular cause of ESKF.¹⁹⁶

Primary MN, comprising 75% of cases, is predominantly associated with M-type phospholipase A2 receptor (PLA2R) as the target antigen, with anti-PLA2R antibodies serving as pathognomonic biomarkers.¹⁹⁷ Approximately 30% of patients achieve spontaneous remission, whereas 30% to 40% progress to ESKF within 10 years. Stratifying patients to tailor treatment strategies remains a challenge, compounded by the fact that 30% to 40% exhibit resistance to current therapies. This underscores the urgent need for the development of novel therapeutic approaches.

EPIDEMIOLOGY

The annual incidence of MN is estimated at 1.2 per 100,000 adults¹⁹⁸ and has remained stable in most ethnic groups, except in the Chinese population, where an increase linked to air pollution exposure is suggested.¹⁹⁹ MN affects all racial groups; however, PLA2R, the predominant target antigen in European and American populations (70%–80% of cases), is detected in only 40% of cases in the Japanese population.²⁰⁰

A peak incidence is observed between 30 and 50 years of age, with a twofold male predominance.²⁰¹ MN is uncommonly seen in children (3%–6% of biopsies),^{202–204} and in this age group it is more likely associated with secondary cause (e.g., hepatitis B).^{205,206}

ETIOLOGY AND PATHOPHYSIOLOGY

MN is an autoimmune, kidney-limited disease driven by autoantibodies targeting specific podocyte antigens, now identifiable in approximately 90% of cases due to groundbreaking advances in research over the past 2 decades.²⁰⁷ Occasionally, in secondary cases, nonpodocyte antigens may be involved. Traditionally, MN is classified into primary (75%) and secondary (25%) forms.

Primary MN is characterized as a renal-limited disease with no identifiable cause, commonly associated with target PLA2R antigen and circulating anti-PLA2R antibodies,²⁰⁸ which are present in 70% to 80% of cases.²⁰⁹ Less commonly, primary MN involves other podocyte antigens including thrombospondin type 1 domain-containing 7A (THSD7A, about 1%–3% of cases),²¹⁰ neural epidermal growth factor-like 1 (NELL1, up to 10% of cases),²¹¹ semaphorin 3B (SEMA3B, 2%, mostly in children),²¹² serine protease HTRA1 (<1%, mainly in elderly),²¹³ and others^{214,215} (Table 30.6).

Secondary MN, in contrast, occurs in association with systemic diseases (e.g., systemic lupus erythematosus); malignancies; infections (e.g., hepatitis B, hepatitis C, and syphilis); medications (e.g., NSAIDs, gold, and penicillamine); or toxins, often involving distinct antigens, rarely PLA2R. For example, secondary MN can be associated with exostosin 1/exostosin 2 (EXT1/EXT2)—seen more commonly in patients with autoimmune disease, particularly lupus nephritis (LN)²¹⁶; protocadherin FAT1, seen in patients with a prior hematopoietic stem cell transplant²¹⁷; neuron-derived neurotrophic factor (NDNF), seen in patients with syphilis²¹⁸; or proprotein convertase subtilisin/kexin type 6 (PCSK6), seen in cases of prolonged NSAIDs use,²¹⁹ among others^{220–222} (see Table 30.6).

MN is driven by an autoimmune process in which circulating antibodies permeate the GBM and, in the subepithelial space, form immune complexes with epitopes on podocyte membranes. Less commonly, MN is linked to exogenous antigens (e.g., hepatitis B) or neoepitopes embedded in the subepithelial zone that serve as targets for circulating antibodies. Alternatively, preformed circulating immune complexes may deposit on the luminal side of the GBM, dissociate, and reassemble in the subepithelial space.

Significant advancements in recent decades have greatly improved the understanding of MN pathogenesis. The Heymann nephritis rat model, which mimics human MN, provided the first major insights into disease pathogenesis. Megalin, a member of the LDL receptor family expressed on rat podocytes, has been identified as the target autoantigen. Circulating autoantibodies that permeate the GBM, bind to megalin on podocytes and form an *in situ* immune complex have been identified as the mechanism of formation of subepithelial deposits seen under the microscope.^{223,224} The first human podocyte antigen discovery was the identification of podocyte neutral endopeptidase (NEP) as an endogenous autoantigen. It has been revealed in a neonatal MN case caused by the transplacental transmission of anti-NEP alloantibodies from a mother with a genetic deletion of NEP, who had been alloimmunized during a previous miscarried pregnancy.²²⁵ However, the most critical milestone came in 2009 with the discovery of PLA2R as a target antigen in approximately 70% to 80% of MN cases, mostly primary.^{208,209} Since then, 13 additional antigens have been identified, linked to both primary and secondary MN, potentially addressing the majority of PLA2R-negative cases.^{207,226}

PLA2R-associated MN represents the predominant form of the disease and is the most extensively studied. PLA2R, a 185-kDa transmembrane glycoprotein, is part of the mannose receptor family and consists of an N-terminal

cysteine-rich region (CysR), a fibronectin-like type II domain, eight C-type lectin domains (CTL1–8), and a C-terminal intracellular tail.^{227,228} This antigen is localized on the cell body and foot processes of human podocytes, though its precise function remains unclear, and it is notably absent in rodent models.²²⁹

The development of glomerular lesions in MN follows a loss of immunologic tolerance to PLA2R antigen, prompting the production of specific autoantibodies. While the causes of tolerance loss are unclear, genome-wide studies have linked a particular HLA-DQA1 haplotype with a 20-fold increased risk in European and Asian populations and certain PLA2R single nucleotide polymorphisms (SNPs) with an approximately fourfold increase.²³⁰ Further studies link MN to other HLA alleles (DQB1 and DRB1)^{231–233} and polymorphisms in immune-regulatory genes like NFKB1 and IRF4,²³⁴ as well as to immune phenotypes marked by reduced regulatory T cells, increased T-helper 17 cells, regulatory B cells, and plasma cells, alongside elevated levels of proinflammatory cytokines such as TNF- α , IL-6, and IL-17.^{235–237} These findings indicate genetic susceptibility related to immune regulation and antigen polymorphism. Although autoantibody triggers remain unknown, environmental factors may play a role—potentially through molecular mimicry between microbial or environmental antigens and PLA2R, as well as prolonged air pollution exposure, potentially explaining rising MN incidence in China.

Anti-PLA2R antibodies in MN predominantly belong to the IgG4 subclass, although IgG1 and IgG3 have occasionally been reported, especially in early disease stages. Autoantibodies target CysR region of the PLA2R protein^{238,239} and may additionally show reactivity to epitopes within CTL1 domains, so-called epitope spreading—a phenomenon related to more severe disease course with higher risk of CKD progression, lower probability of spontaneous remission, or response to rituximab.^{240,241}

Robust evidence suggests a pathogenic role for these antibodies, as studies show a direct correlation between anti-PLA2R antibody titers and their changes over time with the clinical course of the disease.^{208,242–245} Experimental models, such as the Heymann nephritis rat model,^{223,246,247} transgenic mice with human or chimeric PLA2R,^{248,249} and porcine models injected with human anti-PLA2R antibodies,²⁵⁰ replicate MN-like pathology and clinical presentation, underscoring the pathogenicity of these antibodies. Moreover, these models indicate complement activation as a pivotal mechanism of podocyte injury. Complement-dependent cytotoxicity leads to sublytic podocyte injury: actin cytoskeleton disintegration, FPE, and slit-diaphragm disruption,²⁵¹ ultimately resulting in proteinuria; and upregulated expression of genes for the production of oxidants, proteases, prostanooids, growth factors, connective tissue growth factor, TGF, and TGF receptors, leading to the overproduction of extracellular matrix (particularly type IV collagen and laminin), resulting in GBM thickening.^{224,252,253} Of note, altered matrix production may persist after autoantibodies disappear, explaining the delayed resolution of proteinuria despite achieving immunologic remission.²⁵⁴

Renal biopsy findings commonly reveal C3 and C5b-9 (membrane attack complex [MAC]) deposits, emphasizing

Table 30.6 Target Podocyte Antigens in Membranous Nephropathy (MN)

Name	Abbreviation	Year of Discovery	Prevalence (% of MN Cases)	IgG Subclass	Detectable Serum Antibodies	Disease Associations
Mostly Primary MN						
M-type phospholipase A2 receptor	PLA2R	2009	70%-80%	IgG4	Yes	Uncommon, coincidence considered: sarcoidosis, malignancy
Thrombospondin type 1 domain-containing 7A	THSD7A	2014	1%-3%	IgG4	Yes	Malignancy (in 10%-20% of cases)
Neural epidermal growth factor–like 1	NELL1	2019	10%	IgG1	Yes	Malignancy (up to 33% of cases), lipoic acid supplements, mercury (traditional/indigenous medications in India, skin whitening creams)
Semaphorin 3B	SEMA3B	2020	2%	IgG1	Yes	Pediatrics and young adults
Protocadherin 7	PCDH7	2021	2%	IgG1 or IgG4	Yes	Elderly, spontaneous remission
Serine protease HTRA1	HTRA1	2021	<1%	IgG4	Yes	Elderly
Netrin G1	NTNG1	2022	<1%	IgG4	Yes	—
Mostly Secondary MN						
Exostosin 1/Exostosin 2	EXT1/EXT2	2019	7%	IgG1	No	Autoimmune disease (e.g., lupus nephritis class V—associated with favorable kidney survival)
Contactin 1	CNTN1	2020	1%	IgG4	Yes	Chronic inflammatory demyelinating polyneuropathy, male predominance
Neural cell adhesion molecule 1	NCAM1	2021	2%	Variable	Yes	Autoimmune disease (e.g., lupus nephritis class V, in 40% in association to neuropsychiatric SLE)
Transforming growth factor- β receptor 3	TGFBR3	2021	<1%	Missing data	No	Autoimmune disease (e.g., lupus nephritis class V, mostly female)
Protocadherin FAT1	FAT1	2022	1%	IgG4	Yes	Hematopoietic stem cell transplant (MN—the leading glomerular disease after allo-HSCT)
Neuron-derived neurotrophic factor	NDNF	2023	1%	IgG1	Yes	Syphilis
Proprotein convertase subtilisin/kexin type 6	PCSK6	2023	2%	IgG1 or IgG4	Yes	NSAIDs

the role of complement-mediated damage. The lectin pathway appears prominently involved, as evidenced by mannose-binding lectin (MBL) deposition,²⁵⁵ alongside the alternative pathway, supported by positive staining for properdin, complement factor B, and factor H in kidney biopsies.²⁵⁶ In contrast, the classical pathway seems less involved due to IgG4's weak C1q-binding capacity, though IgG1 antibodies may activate this pathway in the early stages of the disease. The complexity of complement engagement in disease pathogenesis remains to be elucidated, particularly in the context of potential targeted therapies.²⁵⁶

NATURAL COURSE OF THE DISEASE

The natural course of MN has been comprehensively summarized by Lerner and colleagues²⁵⁴ and illustrated in Fig. 30.10.

As previously mentioned, the disease begins with the production of anti-PLA2R antibodies (**phase 1: immunologic initiation**), which are initially undetectable in circulation due to full binding to podocyte PLA2R antigen—a phenomenon known as the “sink effect.”²³⁸ Despite this, autoantibodies trigger early histologic changes including immune-complex deposition and podocyte injury, which remain asymptomatic in the early stages of the disease. As disease progresses and antibody production increases, PLA2R binding sites on podocytes become fully saturated,^{209,229} leading to the appearance of anti-PLA2R antibodies in circulation (**phase 2: preclinical seropositivity**). This is accompanied by worsening histologic and clinical features including the onset of subnephrotic proteinuria, often unrecognized or diagnosed incidentally. The precise timing of proteinuria onset remains uncertain. A retrospective study revealed that anti-PLA2R antibodies were detectable more than 1 year before MN diagnosis in 44% of patients and preceded

the first documented proteinuria in 43%. In the remaining cases, autoantibodies appeared concurrently with or after the onset of proteinuria.²⁵⁷ Ongoing autoantibody production and podocyte damage ultimately lead to full-blown nephrotic syndrome (**phase 3: active disease**), with peripheral edema and weight gain prompting medical evaluation and, consequently, MN diagnosis. No specific threshold of anti-PLA2R antibodies correlates with the onset of nephrotic syndrome, and a single serologic measurement is insufficient to predict disease severity or prognosis. Reduction in autoantibody production, either spontaneously or with immunosuppressive therapy, leads to a decline in anti-PLA2R antibodies (**phase 4: immunologic recovery**), followed by serologic remission (**phase 5: immunologic remission**) and a gradual decrease in proteinuria, regression of histologic lesions, and restoration of the GBM. Of note, immunosuppressive regimens differ in their efficacy at suppressing antibody production. Cyclophosphamide combined with glucocorticoids demonstrates the greatest potency, leading to the most rapid decline in anti-PLA2R levels. Rituximab offers an intermediate effect, while calcineurin inhibitors are the slowest to induce immunologic recovery/remission.²⁵⁴

Importantly, immunologic remission precedes clinical remission by several months (3–6 months or longer),²⁴⁵ meaning anti-PLA2R antibodies may become undetectable while proteinuria persists.^{208,242–244} Furthermore, even after the resolution of immune activity, residual proteinuria can persist due to secondary, maladaptive FSGS. Notably, restored GBM and podocytes may remain vulnerable to shear stress, and hyperfiltration (e.g., due to a high-sodium or high-protein diet) may trigger a recurrence of proteinuria independent of MN more readily than in healthy individuals.²⁵⁸ This underscores the importance of continued ACEI/ARB use even after CR.

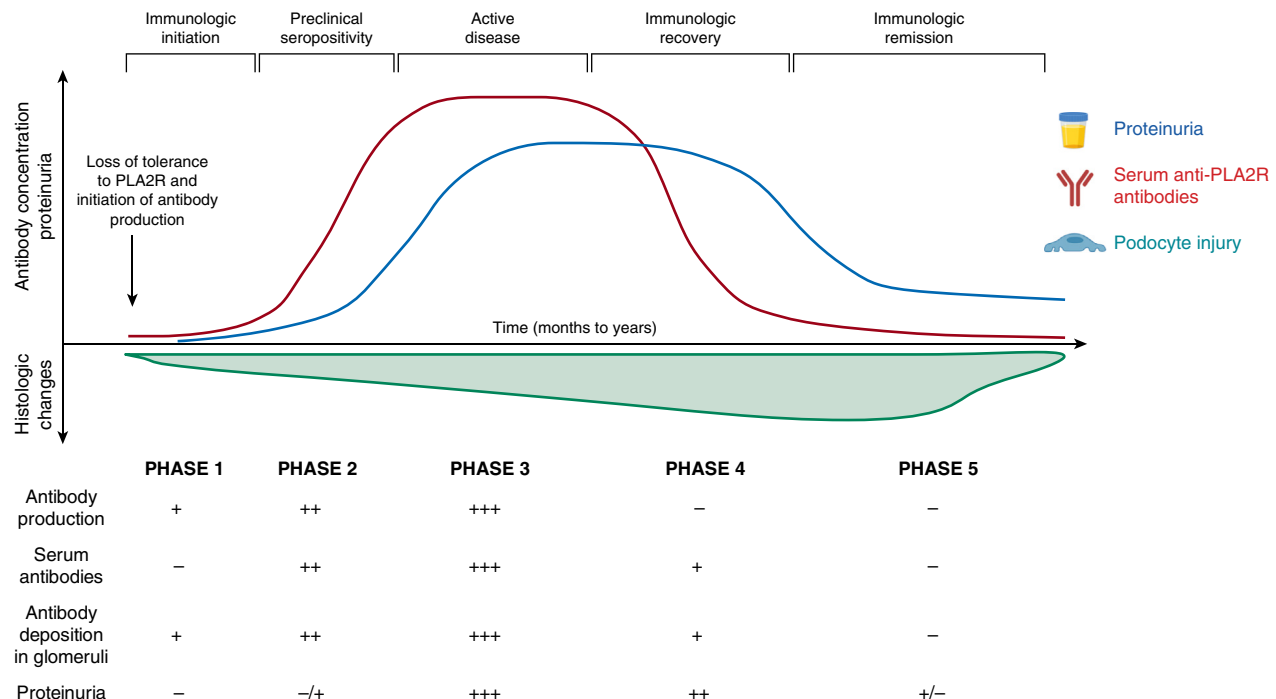


Fig. 30.10 Natural course of membranous nephropathy: immunologic, histologic, and clinical aspects. (Modified from Lerner GB, Virmani S, Henderson JM, Francis JM, Beck LH. A conceptual framework linking immunology, pathology, and clinical features in primary membranous nephropathy. *Kidney Int.* 2021;100:289–300.)

PATHOLOGY

The diagnosis and staging of MN can be reliably established by examining a single glomerulus, as the histologic changes are diffuse and global—uniformly affecting all glomeruli and regions of the glomerular tuft. Although electron microscopy is required to confirm the subepithelial location of deposits, and MN staging was originally defined on the basis of electron microscopy findings, the distinctive features of the disease enable accurate diagnosis and staging using light microscopy and immunofluorescence.

The main histologic feature of MN in light microscopy is diffuse capillary wall thickening without features of inflammation (hypercellularity).²⁵⁹ This finding varies with disease stage and degree of chronic damage. Early-stage lesions (stage I) may be subtle on hematoxylin-eosin stains, while the Jones silver stain highlights basement membrane spikes typically seen in stage II lesions (Fig. 30.11A). In stages III and IV, the GBM appears thickened and irregularly trabeculated. Mesangial hypercellularity is uncommon in primary MN but is more frequent in secondary forms.²⁶⁰ Crescent formation is rare unless there is coexisting anti-GBM disease or ANCA-associated glomerulonephritis.^{261–264} With disease progression, chronic lesions like segmental and global glomerulosclerosis, tubular atrophy, interstitial fibrosis, and mononuclear leukocyte infiltration develop, paralleling the loss of kidney function.²⁵⁹

Immunofluorescence microscopy in MN typically reveals diffuse, finely granular IgG staining along the capillary walls, with IgG4 as the predominant subclass, and complement C3 staining of similar intensity.^{259,265,266} Intense staining for terminal complement components including C5b-9 is often present along capillary walls, though these are not routinely assessed. C1q staining is uncommon in primary MN but appears frequently and intensely in lupus nephritis class V (lupus-associated MN), often alongside IgA and IgM deposits, forming a “full-house” pattern.²⁶⁰ Negative C3 staining is observed in PCDH7-associated MN, whereas segmental or incomplete global IgG staining is seen in the majority of NELL1-associated cases.^{211,267} Tubular basement membrane staining for immunoglobulins or complement is rare in primary MN but common in secondary MN, particularly lupus associated.²⁶⁰

PLA2R antigen staining colocalizing with IgG (or more specifically IgG4) in glomeruli provides higher sensitivity than serum antibody testing (>80%) (see Fig. 30.11B). Most patients with detectable serum anti-PLA2R antibodies also exhibit positive PLA2R antigen staining on biopsy. In 1% to 3% of anti-PLA2R antibody-positive cases, however, PLA2R antigen may be negative on biopsy, likely due to poor epitope accessibility for commercially used reagents or technical artifacts.^{209,268} Conversely, patients who are seronegative for anti-PLA2R antibodies may still show PLA2R antigen positivity in kidney tissue—observed in approximately 70% of seronegative cases.²⁰⁹ This phenomenon, termed the “sink effect,” occurs when antibodies are fully bound to podocyte PLA2R antigen due to their high affinity for it, resulting in undetectable circulating antibody levels.²³⁸ Thus some patients remain seronegative until PLA2R binding sites on podocytes become fully saturated.^{209,229} Secondary MN cases generally lack PLA2R antigen staining; routine staining for novel target antigens in MN has not yet been standardized.

Electron microscopy provides definitive MN diagnosis by revealing diffuse podocyte FPE and subepithelial immune-complex deposits (see Fig. 30.11C), categorized into four ultrastructural stages according to Ehrenreich and Churg’s classification²⁶⁹ (Fig. 30.12):

- **Stage I:** Small electron-dense deposits in the subepithelial zone between the GBM and podocytes, no GBM thickening.
- **Stage II:** Larger deposits interspersed with newly formed GBM material forming “spikes” between deposits; mild GBM thickening.
- **Stage III:** Deposits fully surrounded by new GBM material, showing intramembranous appearance; marked GBM thickening.
- **Stage IV:** Irregular electron-lucent areas within a thickened GBM, resulting from resolution of immune activity and erosion of deposits and loss of their electron density.

CLINICAL PRESENTATION AND DISEASE COURSE

MN typically presents with a nephrotic syndrome of insidious onset and no clear precipitating factor. The onset of the disease is often difficult to identify, as proteinuria develops gradually over months. Biopsy is typically performed after nephrotic syndrome symptoms appear,²⁵⁷ although 20% to 30% of patients are diagnosed while proteinuria remains below 3.5 g/day.²⁷⁰ Hematuria occurs in up to 44%, usually alongside massive proteinuria.¹¹⁴ Kidney function at diagnosis is usually normal or only mildly impaired,²⁷⁰ and AKI is uncommon. Secondary hypertension is infrequent (10%–20%) and typically develops with worsening renal function.

PRIMARY MEMBRANOUS NEPHROPATHY

Among patients with subnephrotic proteinuria (20%–30% of cases), approximately 55% to 60% will progress to nephrotic syndrome,^{270,271} with 70% to 80% progressing within 1 year of diagnosis.^{270,271} The risk of progression is 3.7 times higher in those with anti-PLA2R antibody seropositivity compared with negative serology (HR 3.66, 95% CI: 1.39–9.64),²⁷¹ potentially reflecting distinct disease phases, as proposed by Lerner,²⁵⁴ phase 2 and phase 5, respectively (see earlier). In contrast, 40% do not progress to nephrotic syndrome during long-term follow-up^{270,272} and these patients have an excellent prognosis, with 100% 5-year kidney survival.²⁷⁰

Retrospective data from the Toronto Glomerulonephritis Registry (1974–2003) have shown that 32% of patients with nephrotic syndrome do not respond to treatment.²⁷² Persistent nephrotic syndrome carries a 50% risk of progressing to ESKF within 10 years, compared with 10% with partial remission and almost no risk with CR.²⁷² Risk factors for poor prognosis included older age, male gender, lower baseline eGFR, more severe proteinuria, and poor blood pressure control.²⁷²

Nearly half of patients with partial remission have experienced relapses of nephrotic syndrome, compared with 23% with CR.²⁷² Furthermore, failure to achieve immunologic remission (seronegativity for anti-PLA2R antibodies) is associated with nearly 100% risk of MN relapse.²⁷³

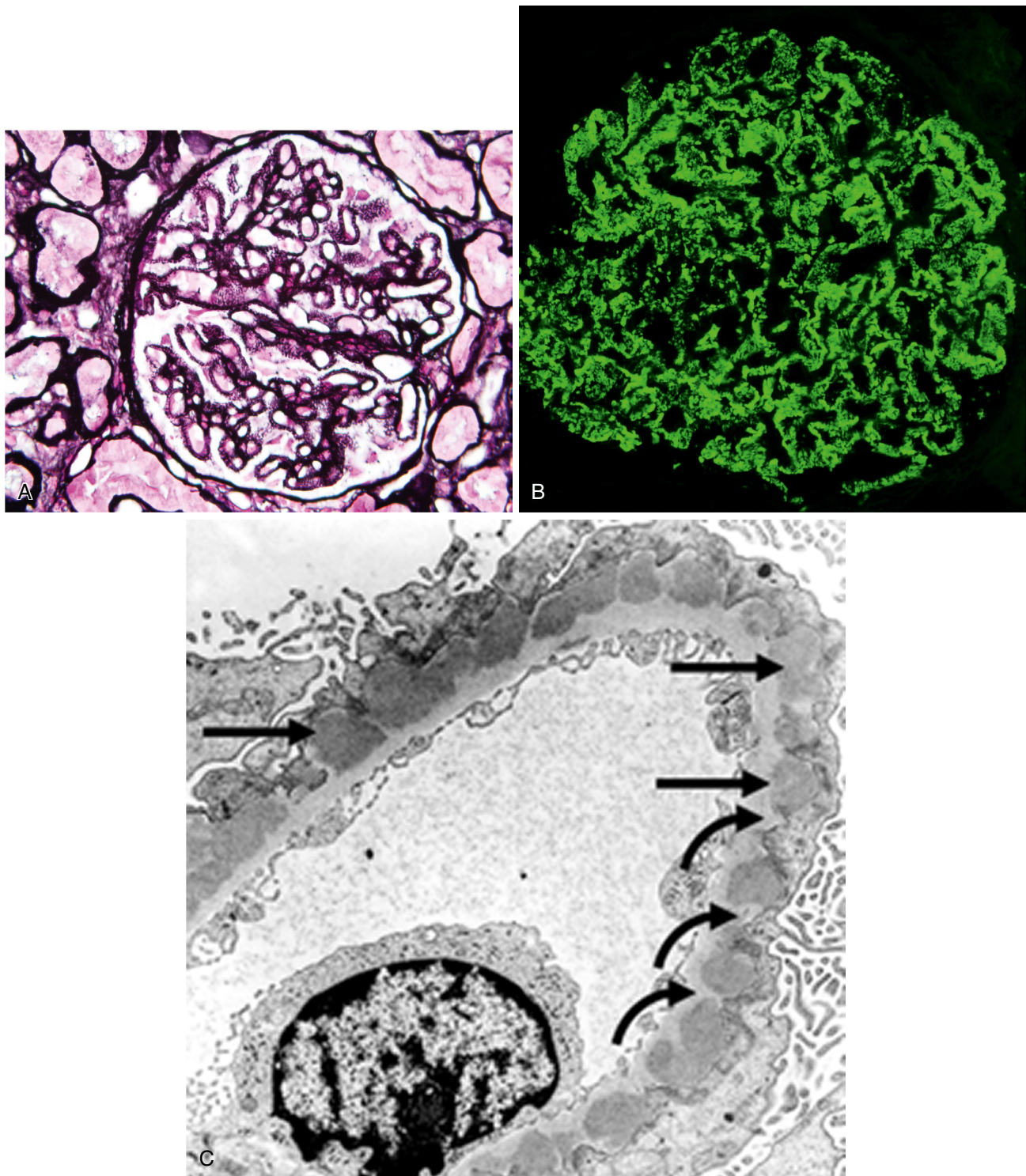


Fig. 30.11 (A) Light micrograph of a glomerulus with features of membranous nephropathy demonstrating spikes, craters, and other irregularities along the outer aspects of the glomerular basement membrane. These correspond to the projections of basement membrane material between and around the immune complex deposits. (Jones methenamine silver stain, $\times 300$.) (B) Immunofluorescence micrograph showing global granular capillary wall staining in a glomerulus with membranous nephropathy representative of PLA2R and also seen with immunoglobulin G in a glomerulus with membranous nephropathy (fluorescein isothiocyanate anti-PLA2R stain, $\times 300$). (A) Electron micrograph showing features of stage II membranous nephropathy with numerous subepithelial dense deposits (*straight arrows*) and adjacent projections of basement membrane material (*curved arrows*) ($\times 100$).

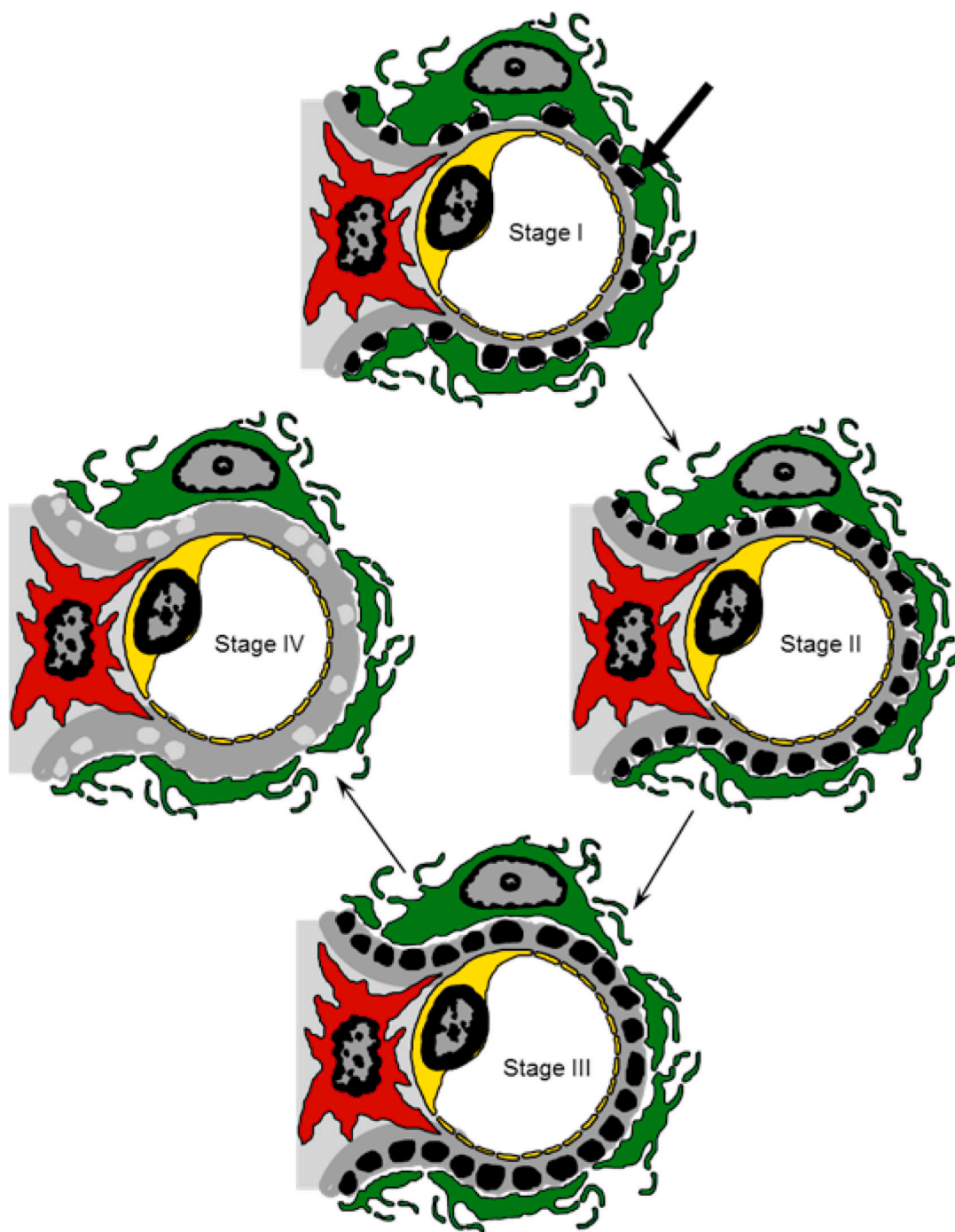


Fig. 30.12 Diagram depicting the four ultrastructural stages of membranous nephropathy. Stage I has subepithelial dense deposits (arrow) without adjacent basement membrane reaction. Stage II has projections of basement membrane adjacent to deposits. Stage III has deposits surrounded by basement membrane. Stage IV has thickened basement membrane with irregular lucent zones. (Courtesy J. C. Jennette.)

SECONDARY MEMBRANOUS NEPHROPATHY

MN co-occurs with cancer in 6% to 22% of cases, representing the most prevalent glomerular disease in patients with malignancies and accounting for approximately 48% to 69% of such cases.^{274,275} The incidence of cancer among patients with MN is reported to be 2.25- to 9.8 times higher than in the general population.^{276,277} In most instances, cancer diagnosis either precedes or is simultaneous with the diagnosis of MN; less commonly, malignancies are identified within a year after the MN diagnosis.^{267,274,277,278} Solid tumors, particularly of

the lungs, prostate, and gastrointestinal tract, are the most frequently associated, primarily in individuals older than 60 years old, suggesting a possible incidental relationship.^{279,280} Antigen-specific associations in malignancy-associated MN have been documented: Malignancy occurs in 33% of NELL1-associated MN,²⁶⁷ in 10% to 20% of THSD7A-associated MN,^{267,281} in 4% to 5% of PLA2R-associated MN,^{267,282} and in 25% of cases with unidentified target antigens.²⁸² THSD7A antigen expression has been detected in tumor tissue from patients with THSD7A-associated

MN, proposing a molecular link in which autoantibodies targeting tumor-expressed THSD7A may cross-react with THSD7A expressed on podocytes *in situ*.^{281,283} In NELL1-associated MN, approximately 75% of patients experience remission of proteinuria in response to cancer-directed therapy alone, suggesting a direct pathogenic relationship between malignancy and MN in these cases.²⁶⁷ By contrast, PLA2R-associated MN shows no significant clinical response to cancer treatment, implying that malignancies in this subgroup are likely incidental²⁶⁷; however, a single case has reported PLA2R expression within tumor tissue.²⁸⁴

MN is the leading glomerular disease in patients after allo-HSCT, accounting for two-thirds of cases. The development of MN is usually associated with a decrease in immunosuppression and graft-versus-host disease (GvHD)²⁸⁵ and improved after increasing in immunosuppression. These cases have been found to be related to FAT1 antigen in most cases.²¹⁷ For further discussion on cancers and kidney disease see [Chapter 57](#).

The autoimmune disease presenting most commonly with MN is systemic lupus erythematosus (SLE). Around 10% to 20% of patients present with class V lupus nephritis (LN), corresponding to the MN pattern of injury. These patients are primarily young women, presenting with NS, often without other symptoms of SLE. Several antigens are related to class V LN including EXT-1/EXT-2 and, less frequently, NCAM1 or TGFBR3.^{221,222,286} Kidney outcome is favorable. For further discussion on lupus, see [Chapter 31](#). MN has also been linked to sarcoidosis including both PLA2R-positive and negative cases, though a causal relationship remains uncertain.

Several drugs and toxins have been associated with MN including gold, penicillamine, anti-TNF agents (adalimumab, etanercept, infliximab), lipoic acid,^{287,288} and mercury-contaminated products such as indigenous medicines and skin-whitening creams.^{289,290} While the pathomechanisms of drug-induced MN remain unclear, NELL1-associated MN has been identified in cases linked to lipoic acid and mercury exposure.

Lastly, MN has been linked to infections, particularly hepatitis B, and less commonly hepatitis C or HIV, all of which should be excluded during diagnosis. Syphilis has also been associated with MN lesions, with NDNF identified as a target antigen.²¹⁸ The clinical course of secondary MN varies depending on the underlying cause and effectiveness of its treatment.

DIAGNOSTIC APPROACH

The diagnostic algorithm for nephrotic syndrome in adults with a clinical course suggestive of MN includes anti-PLA2R antibody testing; kidney biopsy (except in certain cases, see later); and exclusion of secondary causes of glomerular disease.

Anti-PLA2R antibody testing, using either ELISA or indirect immunofluorescence assay (IIF), should be performed in all cases. Both methods, FDA approved since 2014, offer near-perfect specificity (96%–100%) and sensitivity ranging from 64% to 87%,²⁹¹ likely reflecting variation across different immunologic phases of the disease at testing. ELISA, although slightly less sensitive, provides

quantitative results, enabling monitoring of disease activity, prognostication, and guiding clinical decision making. A universally accepted cutoff has not been established, with thresholds ranging from 2 to 20 RU/mL across studies. Uncertain results should be confirmed using the more sensitive qualitative or semiquantitative IIF assay.²⁶⁸

Kidney biopsy is the gold standard for diagnosing glomerular diseases including MN. However, the accuracy and availability of anti-PLA2R antibody testing allow for noninvasive diagnosis in patients with a typical presentation, preserved kidney function, and no MN-associated conditions (e.g., lupus, hepatitis, and malignancy) or diabetes.^{40,268,292} Studies show that additional findings on biopsy are seen in only 0% to 3.3% of such cases, none affecting clinical management. In contrast, 8.1% to 23% of patients with eGFR <60 mL/min/1.73 m² have concurrent diagnoses, highlighting the need for biopsy in those with impaired kidney function.^{40,293} A biopsy is recommended in cases of atypical presentations (e.g., rapid eGFR decline and positive antinuclear antibodies), concurrent diabetes, lack of therapeutic response, or anti-PLA2R seronegativity (as positive PLA2R staining is observed in 70% of such cases²⁰⁹).

In all MN cases, an evaluation for secondary conditions is recommended, including a detailed patient history (e.g., drug use and systemic diseases), physical examination, viral serology, antinuclear antibodies, and age- and sex-appropriate screening for malignancy ([Fig. 30.13](#)). Special attention should be given to cancer screening in PLA2R-negative and THSD7A- and NELL1-associated MN. Although PLA2R-associated MN is typically primary, coexisting conditions have been reported in 6% to 16% to 29% of cases including malignancies (23%–43%), paraproteinemias (2%–17%), autoimmune diseases (13%–31%, particularly sarcoidosis), and hepatitis (13%).^{268,282,292,293} While it remains unclear whether these conditions are related to PLA2R-associated MN or coincidental, their considerable prevalence warrants a thorough workup.

TREATMENT AND MONITORING

Primary MN treatment involves conservative therapy and, for patients at risk of progressive kidney injury, immunosuppressive therapy. A key element of conservative care is the use of ACEi or ARB at the maximum tolerated doses. These agents reduce proteinuria and provide nephroprotection, but they also increase the likelihood of spontaneous remission. Supportive therapy includes symptomatic management (e.g., edema control, lipid-lowering medications, and thromboprophylaxis) and lifestyle modifications (outlined earlier and in [Fig. 30.2](#)). Decisions regarding immunosuppressive therapy are based on a stratified assessment of ESKF risk, with the goal of providing effective treatment to patients at high risk while avoiding unnecessary immunosuppression in those with a favorable prognosis and a high likelihood of spontaneous remission.

Treatment of secondary forms of MN should be aimed at the underlying systemic disease or etiology (e.g., resection of a tumor in malignancy-associated MN and treatment of hepatitis B in viral-associated MN). Immunosuppressive therapy is considered only in cases of treatment failure or severe clinical presentations.

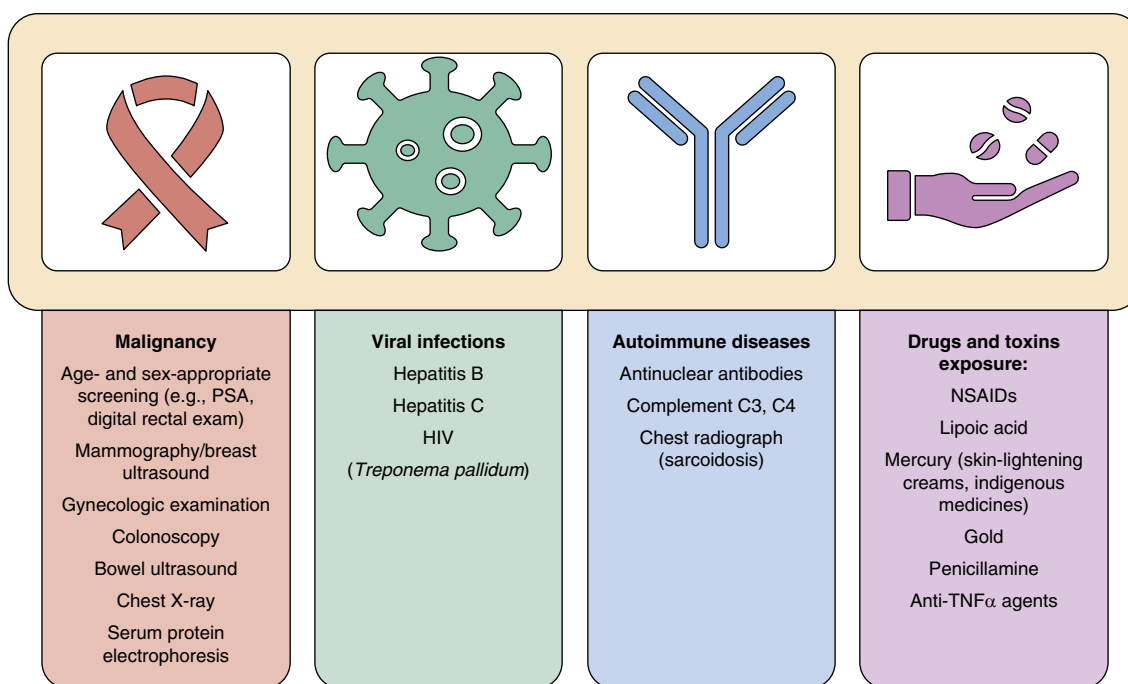


Fig. 30.13 Diagnostic evaluation for secondary causes of membranous nephropathy.

INITIAL THERAPEUTIC APPROACH IN HIGH-RISK PRIMARY MEMBRANOUS NEPHROPATHY

At diagnosis, very high-risk patients should be identified promptly and started on immunosuppressive therapy alongside conservative treatment without delay (Fig. 30.14). This group includes patients with rapidly declining kidney function due to MN or those with severe, complicated nephrotic syndrome. The preferred regimen is alternate therapy with an alkylating agent (cyclophosphamide is favored over chlorambucil due to its superior safety profile) and glucocorticoids, the only treatment shown to improve both kidney and patient survival.

For patients not classified as very high risk, a 6-month trial of supportive therapy is recommended. This includes gradually increasing ACEi/ARB doses to the maximum tolerated levels. However, titration can be challenging, as most patients are normotensive, and higher doses of ACEi/ARB may not be well tolerated.

MONITORING DISEASE PROGRESSION

Regular monitoring, at intervals of at least 2 to 3 months, is crucial and should assess both the clinical course (kidney function, proteinuria, serum albumin) and immunologic activity (anti-PLA2R antibody titers measured via ELISA).

The current risk stratification algorithm also incorporates the evaluation of IgG excretion (either 24-hour measurement or selectivity index calculation) and low-molecular-weight protein excretion, α 1-microglobulin, β 2-microglobulin, as biomarkers. However, these tests are rarely performed in routine clinical practice. In particular, β 2-microglobulin assessment is technically demanding, requiring urine alkalization and highly sensitive detection methods.

THERAPEUTIC APPROACH AFTER 6 MONTHS OF CONSERVATIVE CARE

After 6 months of conservative therapy, patients should be reassessed for ESKF risk and indications for immunosuppression (Table 30.7, see Fig. 30.14). Immunosuppressive therapy should be initiated earlier in cases of a worsening clinical course (e.g., rising proteinuria, declining kidney function, and nephrotic syndrome complications) or persistent immunologic activity, with treatment tailored to the specific clinical scenario.²⁹⁴

Patients with normal kidney function and proteinuria <3.5 g/day or nephrotic proteinuria showing a $>50\%$ decrease during the 6-month supportive therapy are considered low risk for ESKF progression, with a favorable long-term prognosis. Immunosuppressive therapy does not appear to offer additional benefits for these patients while posing a risk of adverse effects. This group should remain monitored and continue supportive therapy. Adding an SGLT2 inhibitor seems reasonable, given its additional antiproteinuric effects in glomerular diseases.

Patients with preserved eGFR but persistent nephrotic syndrome and $<50\%$ reduction in proteinuria during ACEi/ARB treatment are classified as a moderate-risk group. Therapeutic decisions for this group should be individualized including options such as continued conservative therapy, monotherapy with CNI, combined CNI and glucocorticoid therapy, or rituximab.

Patients with impaired kidney function (eGFR <60 mL/min/1.73 m²), massive nephrotic syndrome despite 6 months of conservative therapy (proteinuria >8 g/day), or nephrotic proteinuria with additional risk factors (at least one of the following: serum albumin <2.5 g/dL, persistent PLA2R >50 RU/mL, high excretion of

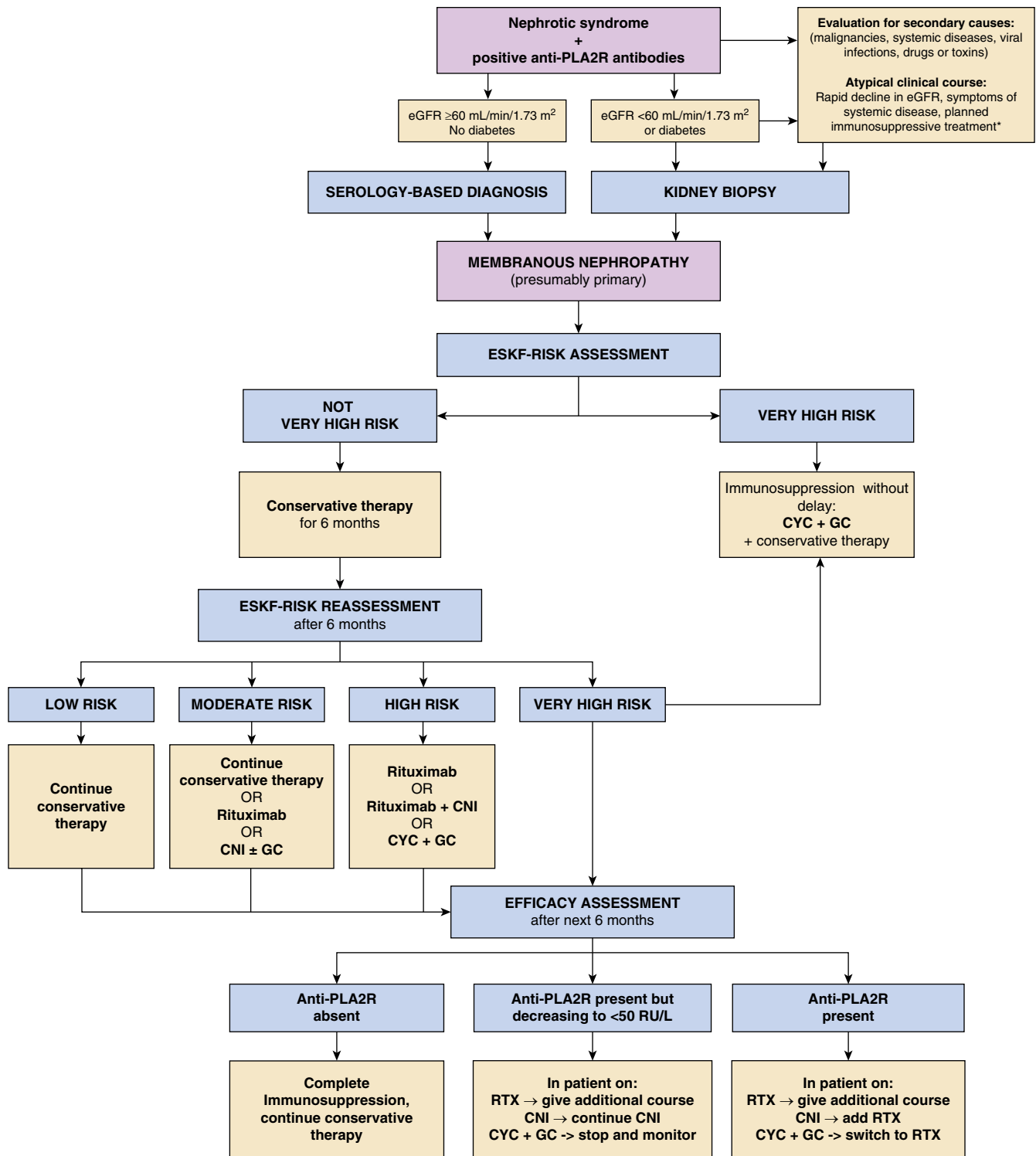


Fig. 30.14 Diagnostic-therapeutic algorithm for membranous nephropathy.

low-molecular-weight proteins or IgG) are considered high risk for kidney failure. For these patients, monotherapy with CNI is not recommended due to inferior outcomes compared with rituximab and a high relapse rate after drug withdrawal. Preferred therapeutic options include rituximab monotherapy, rituximab with CNI, or, alternatively, cyclophosphamide + glucocorticoids.

Decisions should be based on individual risk profiles, patient preferences, and local health care system factors (e.g., availability and cost coverage). Cyclophosphamide + glucocorticoids, with its less favorable toxicity profile, should be reserved for very high and selected high-risk patients.

In cases of immunologic remission (absence of anti-PLA2R antibodies) after 6 months of immunosuppressive

Table 30.7 Risk Stratification for End-Stage Kidney Failure in Membranous Nephropathy and Suggested Choice of Immunosuppressive Treatment

Risk Class	Description	Treatment Choice	Drug Dosing
Very high	Assessed at diagnosis or at any point during follow-up: Rapid deterioration of kidney function; OR Life-threatening nephrotic syndrome	Cyclophosphamide + Glucocorticoids	Cyclophosphamide + Glucocorticoids – cyclical regimen: GC—months 1, 3, and 5 Methylprednisolone 1 g i.v. for 3 consecutive days, followed by prednisone 0.5 mg/kg/day orally
High	Assessed after 6 months of conservative management: eGFR <60 mL/min/1.73 m ² ; OR Proteinuria >8 g/day for 6 months; OR Proteinuria >3.5 g/day and no decrease >50% after 6 months of conservative therapy AND at least 1 of the following: Serum albumin <2.5 g/dL, Anti-PLA2R antibodies >50 RU/mL, Urinary α 1-microglobulin >40 μ g/min, Urinary β 2-microglobulin >1 μ g/min, Urinary IgG >250 mg/day Selectivity index >0.2	Rituximab OR Rituximab + Calcineurin inhibitor OR Cyclophosphamide + Glucocorticoids	AND CYC—months 2, 4, and 6: 2.5 mg/kg/day orally Rituximab: 2x 1.0 g i.v. 2 weeks apart OR 4x 375 mg/m ² i.v. at weekly intervals Calcineurin inhibitors: Tacrolimus for 12 months 0.05–0.1 mg/kg/day, Target trough level: 3–8 ng/mL OR Cyclosporin for 12 months 3.5 mg/kg/day, Target trough level: 125–225 ng/mL
Moderate	Assessed after 6 months of conservative management: eGFR $\geq$ 60 mL/min/1.73 m ² ; AND Proteinuria >3.5 g/day AND no decrease >50% after 6 months of conservative therapy; AND No criteria for high risk met	Conservative therapy OR Rituximab OR Calcineurin inhibitor $\pm$ Glucocorticoids	
Low	Assessed after 6 months of conservative management: eGFR $\geq$ 60 mL/min/1.73m ² ; AND Proteinuria <3.5 g/day or a decrease >50% in 6 months of conservative therapy	Conservative therapy	

treatment, therapy should be discontinued. If there is immunologic recovery but no remission (low-level, decreasing anti-PLA2R antibodies), immunosuppressive therapy should continue, except for cyclophosphamide to avoid high cumulative doses and related toxicity (>10 g, potentially affecting fertility, >36 g, increasing the risk of malignancy). These patients should be closely monitored.

REFRACTORY DISEASE

Patients who fail to achieve immunologic recovery after 6 months of immunosuppressive therapy require intensified treatment. For worsening kidney function, cyclophosphamide-based regimens are preferred, while rituximab-based therapies are used in other cases. Combined regimens with cyclophosphamide + rituximab + glucocorticoids also appear promising, showing high efficacy. Experimental treatments for refractory cases include therapies targeting B cells (e.g., obinutuzumab—anti-CD20 antibody; belimumab—anti-BLYSS antibody), plasma cells (e.g., bortezomib—proteasome inhibitor; daratumumab, felzartamab—anti-CD38 antibodies), or the complement system (e.g., narsoplimab—MASP2 inhibitor, iptacopan—factor B inhibitor).

RELAPSES

Relapses of MN are typically managed with rituximab (alone or with CNI). For high-risk patients with progressive kidney function decline, cyclophosphamide + glucocorticoids should be preferred.

OUTCOMES

KIDNEY OUTCOMES

Despite a reported 10-year renal survival rate of 60% to 80%,^{295–297} MN remains one of the leading glomerular diseases causing ESKF.¹⁹⁶ Importantly, progression to ESKF depends strongly on remission of proteinuria: the 10-year risk in patients who undergo complete or partial remission of proteinuria ranges from 0% to 10%, compared with a >50% incidence of ESKF in patients who maintain nephrotic-range proteinuria.²⁷²

POST-TRANSPLANT MEMBRANOUS NEPHROPATHY RECURRENCE

The 10-year risk of recurrent MN following kidney transplantation is considerable, estimated at 31%,²⁹⁸ and increases to 42% when routine protocol biopsies of the graft are performed.²⁹⁹ The main risk factor is the presence of anti-PLA2R antibodies at the time of transplantation,^{298,300}

which is associated with a >18-fold increase in recurrence risk (HR: 18.58; 95% CI, 5.37–64.27).²⁹⁸ MN recurs within 2 years post transplant in approximately half of patients, with significantly earlier recurrence observed in individuals with high pretransplant anti-PLA2R antibody titers (>150 RU/mL) (median time to recurrence: 7 months vs. 21 months).²⁹⁸ Although evidence is limited, preemptive rituximab therapy may be beneficial in patients with high pretransplant anti-PLA2R antibody levels, severe proteinuria, and/or a history of graft loss due to MN recurrence, particularly if living-donor transplantation is planned.³⁰¹ Alternatively, close serologic monitoring of anti-PLA2R antibody levels is required, with testing every 1 to 3 months for the first 6 to 12 months post transplant, followed by every 3 to 6 months thereafter depending on PLA2R status. In cases of biopsy-confirmed MN recurrence with proteinuria exceeding 1 g/day, rituximab therapy is recommended,⁴⁰ resulting in a 68% rate of partial or CR and, consequently, improved graft survival.²⁹⁸

Conversely, *de novo* cases of MN in the allograft kidney tend not to be PLA2R positive and have been associated with concurrent antibody-mediated rejection and, subsequently, reduced graft survival.³⁰²

NEPHROTIC SYNDROME IN OTHER GLOMERULAR AND SYSTEMIC DISEASES

GLOMERULONEPHRITIDES WITH NEPHROTIC SYNDROME

Immune-complex (e.g., LN), pauci-immune (e.g., ANCA-associated glomerulonephritis), and linear (e.g., anti-GBM disease) forms of glomerulonephritis typically present with a nephritic picture that is marked by microscopic hematuria, subnephrotic proteinuria, and various degrees of AKI. Nephrotic-range proteinuria is unusual in rapidly progressive forms of glomerulonephritis despite extensive injury to the podocyte and filtration barrier in part due to reduced filtration of protein loads. In contrast, more chronic forms of glomerulonephritis including C3 glomerulopathy, immune complex-mediated membranoproliferative glomerulonephritis, LN, and IgA nephropathy can occasionally present with nephrotic-range proteinuria and, in some instances, full nephrotic syndrome.

Nephrotic syndrome in LN patients can be seen in all classes but is most commonly seen in membranous LN, class V, and the much rarer entity of “lupus podocytopathies,” in which >90% of patients meet all criteria for nephrotic syndrome with a clinical course more typical of MCD or FSGS. Lupus podocytopathy is diagnosed by kidney biopsy showing normal light microscopy (akin to minimal change disease) or FSGS lesions with either no or mesangial-limited proliferation, immunofluorescence with no more than mesangial deposits (i.e., no glomerular capillary wall deposits), and electron microscopy with extensive FPE (and again no more than mesangial deposits). The presence of severe FPE in the absence of deposits in the GBM implies a mechanism independent of immune-complex deposition and likely similar to MCD or primary FSGS, with production of a cytokine or lymphokine toxic to podocytes driven by T-cell dysfunction.⁸¹

A similar “podocytopathy” has been described in IgA nephropathy, sometimes termed IgA nephropathy but more

commonly described as concomitant IgA nephropathy with MCD.³⁰³ As with lupus podocytopathy, the hallmark feature is extensive FPE not typical for IgA nephropathy, with light microscopy and immunofluorescence showing features usually seen in mild forms of the disease. The clinical course usually follows that seen in MCD, with high rates of steroid responsiveness coupled with frequent relapses of nephrotic syndrome requiring steroid-sparing, alternative therapies.

AMYLOIDOSIS

Amyloidosis comprises a diverse group of systemic and local diseases characterized by the extracellular deposition of characteristic fibrils in various organs. More than 25 different proteins can produce amyloid fibrils, all of which bind Congo red (Fig. 30.15A), leading to diagnostic apple green birefringence under polarized light and thioflavin T. Amyloid fibrils have a characteristic ultrastructural appearance with randomly oriented, 8- to 12-nm nonbranching fibrils (see Fig. 30.15B).³⁰⁴

Only some amyloid proteins deposit in the kidney. Most renal amyloidoses are due to either AL amyloidosis or AA amyloidosis. In one series, 1.3% of 21,500 renal biopsies over an 8.5-year period were found to have amyloidosis with 86% AL amyloid, 7% AA amyloid, and the remainder being a variety of types of hereditary amyloids. In another single-center series of 474 amyloidosis patients evaluated from 2007 to 2011, 85% were AL amyloid, 7% AA amyloid, and 4% hereditary amyloid. In AL amyloidosis, the deposited fibrils are derived from the variable portion of immunoglobulin light chains produced by a clonal population of plasma cells or plasmacytic B cells. AA amyloid results from the deposition of serum amyloid A (SAA) protein in chronic inflammatory states.^{304,305} While in Westernized countries, AA amyloid is most commonly found in association with rheumatoid arthritis and other inflammatory arthritides, it is also seen with inflammatory bowel disease, familial Mediterranean fever (FMF), patients with spinal cord injury with chronic urinary infections and decubitus ulcers, bronchiectasis, poorly treated osteomyelitis, and in persons addicted to heroin or other agents who inject drugs subcutaneously.

A small fraction of renal amyloidoses is due to rare hereditary forms of amyloidosis, such as those caused by inherited gene mutations encoding transthyretin (ATTR), fibrinogen A α -chain (AFib), apolipoprotein A-I (apo A-I) or A-II (apo A-II), lysozyme (ALys), gelsolin (AGel), and leukocyte chemotactic factor 2 (LECT2) peptide. Renal amyloidosis caused by LECT2 defects is a more common cause of amyloidosis than AL amyloid in some populations (e.g., Latinos from the southwest United States, Punjabis, Native Americans, and Egyptians). Laser microdissection and mass spectrometry are necessary to determine the origin of some of these unusual amyloid subtypes.³⁰⁶

In AL amyloidosis, fibrils are composed of the N-terminal amino acid residues of the variable region of an immunoglobulin light chain. λ -Light chains predominate over κ -light chains, and there is an increased incidence of monoclonal λ subtype VI. The diagnosis of AL amyloidosis may be suspected on clinical grounds but requires biopsy documentation.³⁰⁷ The kidneys are the most common major organ involved in AL amyloid, and most patients eventually develop renal amyloid. In the past, up to 10% to

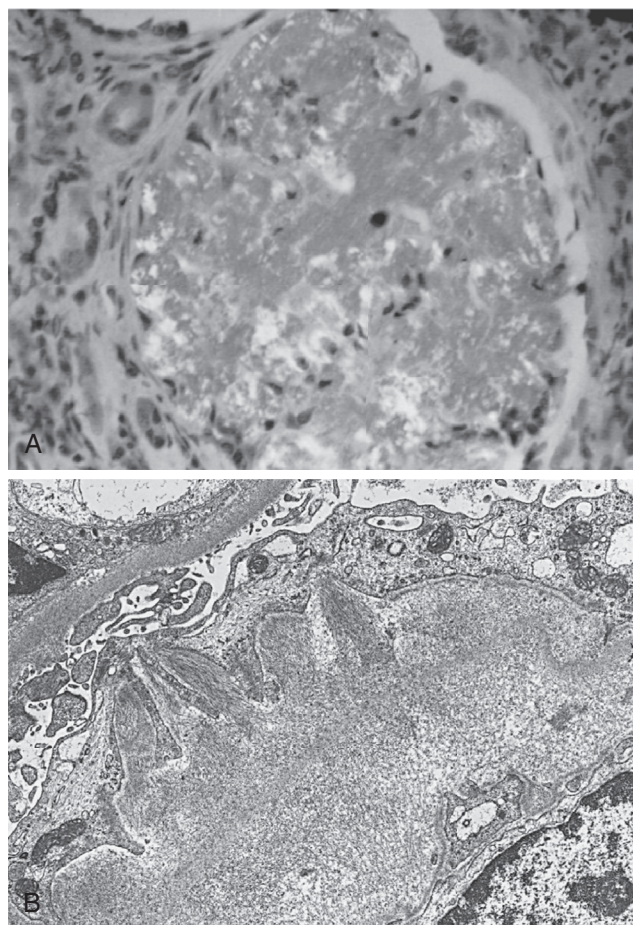


Fig. 30.15 Amyloidosis. (A) Congo red stain of a glomerulus that is largely replaced by amyloid demonstrates the characteristic birefringence under polarized light ($\times 450$). (B) Amyloidosis. Electron micrograph showing extensive infiltration of the glomerular basement membrane by 10-nm fibrils that project toward the urinary space ($\times 8000$).

20% of patients older than 60 years with presumed INS had amyloidosis on renal biopsy. Multiple myeloma occurs in up to 20% of AL amyloidosis cases. Amyloidosis should be suspected in all patients with circulating serum monoclonal M proteins, and approximately 90% of primary amyloid patients will have a paraprotein spike in the serum or urine by immunofixation.^{304,305} A greater percentage, on free light chain assay, will have an elevation and predominance of one circulating light chain.

The incidence of AL amyloid is about 8 per million annually. Most patients with AL amyloidosis are older than 50 years (median age 59–63 years), and <1% are younger than 40 years of age. Men are affected twice as often as women. Presenting symptoms include weight loss, fatigue, light-headedness, shortness of breath, peripheral edema, pain due to peripheral neuropathy (often carpal tunnel syndrome), and orthostatic hypotension. Patients may have cardiomyopathy, hepatosplenomegaly, macroglossia, or rarely enlarged lymph nodes. Multisystem organ involvement is typical, with commonly affected organs being the kidney (50%), heart (40%), and peripheral nerves (up to 25%).³⁰⁷ In one series, 25% of AL amyloid patients had one major

organ system involved, 36% had two organ systems involved, and 38% had three or more involved.³⁰⁸

The diagnosis of amyloid is usually established by tissue biopsy of an affected organ.^{309–311} Liver and kidney biopsies are positive in up to 90% of clinically affected cases. A diagnosis may be made less invasively with fat pad aspirate (60%–90%), rectal biopsy (50%–80%), bone marrow aspirate (30%–50%), gingival biopsy (60%), or dermal biopsy (50%).³¹² Serum amyloid P (SAP) whole-body scintigraphy following injection of radiolabeled SAP allows the noninvasive diagnosis of amyloidosis, as well as a quantification of the extent of organ system involvement and assessment of the response to treatment.³¹³ This test may be positive even when tissue biopsy has been negative and may be more accurate in AA than AL amyloidosis. In AL amyloidosis, detection of an abnormal ratio of free κ - to λ -light chains in the serum suggests plasma cell dyscrasias with a higher sensitivity than either serum or urinary protein electrophoresis and immunofixation.³¹⁴ It also allows assessment of response to therapy by following the level of abnormal free light chain in the serum.³¹⁵

Patients with hereditary amyloidosis due to deposition of abnormal transthyretin, apolipoproteins, lysozyme, or fibrinogen Aa may present in a fashion similar to AL amyloid. In one series, 10% of 350 cases of hereditary amyloidosis were misdiagnosed as having AL amyloid.³¹⁶ Although hereditary amyloidosis may present at any age, most patients are adults and present with higher serum creatinine and less proteinuria than those with AL or AA amyloidosis. Some hereditary types lead to predominant amyloid deposition in the tubulointerstitial and medullary compartment (e.g., amyloid due to apo A-I, A-II, and A-IV). Other forms of hereditary amyloidosis are associated with massive glomerular obliteration (AFib).³¹⁷ LECT2 fibrils deposit in all compartments of the kidney and can lead to progressive kidney failure. Establishing the correct diagnosis is crucial because the treatment of hereditary amyloid may include liver transplantation rather than chemotherapy or stem cell transplantation as in AL amyloid. Laser dissection of involved glomeruli and proteomic analysis by mass spectrometry, a test currently available in few specialized centers, can accurately diagnose all types of glomerular amyloid.³¹⁸

The optimal treatment for AL amyloid depends on the patient's age, organ system involvement, and overall health.³¹⁹ In general, healthier and younger patients have been offered autologous stem cell transplants (ASCT), whereas older patients with more morbid-associated conditions have received chemotherapeutic regimens. For patients presenting with a very low GFR (<20 mL/min/1.73 m²), kidney survival depends on the magnitude and speed of the response of the hematologic disorder to treatment. Regardless of whether chemotherapy or stem cell transplant is used, the treatment of nephrotic amyloid patients requires supportive care. This may include judicious use of diuretics and salt restriction in those with edema and treatment of orthostatic hypotension with compression stockings, fludrocortisone, and midodrine.

The treatment of AA amyloid focuses on the control of the underlying inflammatory disease process. This has included surgical debridement of inflammatory tissue,

antibiotic therapy of infectious processes, antiinflammatory and immunosuppressive agents (including newer biologics) in rheumatoid arthritis and inflammatory bowel disease, and chronic colchicine therapy in FMF. Therapy may lead to stabilization of renal function, reduction in proteinuria, and resolution of amyloid deposits. Prognosis may be good if the underlying disease can be controlled, and there is not already extensive amyloid deposition.^{320,321}

Experience with renal transplantation in AL amyloid is limited, but transplantation may be performed either before or after ASCT.³²² The 5- and 10-year survival rates of renal transplant patients with AA amyloid, largely FMF patients, who received living-related allografts have been as high as 80% and 66%, respectively. In hereditary amyloidosis due to fibrinogen A α -chain disease, recurrence occurred in 50% of allografts with frequent graft loss. Better results are reported with combined liver and kidney transplantation.

FIBRILLARY GLOMERULONEPHRITIS AND IMMUNOTACTOID GLOMERULONEPHRITIS

Fibrillary glomerulonephritis (FGN), a rare disease with incidence of <1% of native kidney biopsies, was first described in 1977. Clinically, the patient with FGN can present with hypertension, proteinuria (full nephrotic syndrome in up to 50% of patients), hematuria, and renal insufficiency. Most patients present in the fifth or sixth decade of life, and >90% of patients in the two largest series of FGN self-identified as white.

Currently, a diagnosis of FGN can only be made via kidney biopsy. On light microscopy, the patterns of injury include mesangioproliferative, membranoproliferative, diffuse proliferative, membranous, diffuse sclerosing, and crescentic forms of glomerulonephritis, although the crescentic variant is only rarely encountered. Congo red staining is negative. On immunofluorescence, IgG (most cases are polyclonal), kappa and lambda light chains, and complement are present and often have a “smudgy” appearance. On electron microscopy, randomly arranged fibrillary deposits, 16 to 24 nm in diameter (larger than amyloid), are seen in the mesangium and capillary walls (Fig. 30.16).^{323–325}

Laser microdissection and mass spectrometry have provided further information on the composition of the classic fibrillary deposits of FGN. This process involves the extraction of glomerular protein, digestion, and identification of the protein constituents via amino acid sequencing. Apoprotein E, serum amyloid P, and immunoglobulin heavy/chain C region are some of the identifiable proteins that can help distinguish amyloidosis from FGN from ITGN.³²⁶ The discovery of DNAJB9 in the glomeruli of FGN patients by mass spectrometry and use of anti-DNAJB9 antibodies for immunofluorescence/immunohistochemistry has transformed the diagnosis of this disease, as the presence of DNAJB9 staining has now become a more definitive way to diagnose FGN than EM appearance. DNAJB9 is a member of a family of cochaperones to heat shock proteins, which play an important role in the proper folding/unfolding/translocation/degradation of key cellular protein constituents.^{327–331}

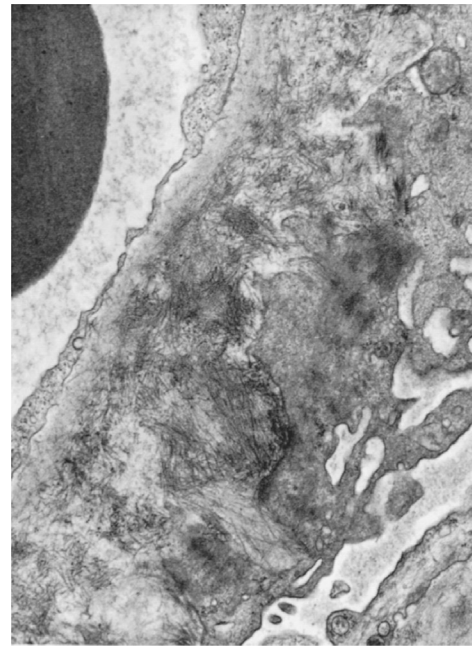


Fig. 30.16 Fibrillary glomerulonephritis. Electron micrograph showing the characteristic randomly oriented fibrils, measuring 16 to 20 nm within the glomerular basement membrane. The foot processes are effaced (x8000).

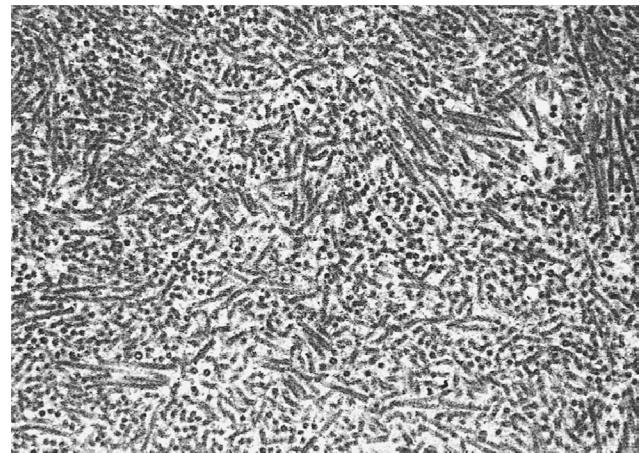


Fig. 30.17 Immunotactoid glomerulonephritis.

Most FGN cases at this time are considered primary. In a series of 66 FGN cases from the Mayo clinic, 15 (23%) were associated with malignancy (mostly carcinomas, including 6 with multiple myeloma). Ten (15%) were associated with autoimmune disease (Crohn, SLE, Graves disease, ITP, PBC, ankylosing spondylitis, scleroderma, and Sjogren). Two patients (3%) were hepatitis C positive.³²⁵ An older series of 42 FGN patients (26% black) from the University of North Carolina included 7/26 (27%) testing positive for hepatitis C with no detectable cryoglobulinemia. These patients had the poorest outcome, with all but one proceeding to ESRD, although these cases predate the era of effective antiviral therapy.³³² Aside from treating an underlying disorder in secondary cases, immunosuppressive agents including corticosteroids, cyclophosphamide, mycophenolate,

calcineurin inhibitors, azathioprine, and rituximab have been used with limited success.³³³ The reported incidence of FGN recurrence in the allograft is varied, ranging from 8% to 36% across small series.³³⁴

Immunotactoid glomerulonephritis (ITGN) is 10-fold less common than FGN, representing only 0.06% of all native kidney biopsies. Patients with ITGN can present with proteinuria (full nephrotic syndrome in up to 70%), microscopic hematuria, renal insufficiency, and hypertension. The average age of presentation is 60 years.

The diagnosis of ITGN requires a kidney biopsy. On light microscopy, the patterns of injury seen include membranoproliferative, diffuse proliferative, and membranous glomerulonephritis. Immunofluorescence staining shows the presence of IgG or IgM (often monoclonal), kappa and lambda light chains, and complement. On electron microscopy, there are glomerular microtubules measuring 30 to 50 nm in diameter (larger than FGN or amyloidosis) in parallel arrangement, located predominantly in the subepithelial and subendothelial space (Fig. 30.17). Cryoglobulin deposits on EM can be indistinguishable from ITGN and must be excluded clinically.³³⁵

The cause of ITGN is unclear. It was once thought that FGN and ITGN were related entities. But compared with FGN, there is greater likelihood of hypocomplementemia, monoclonal gammopathy, and lymphoproliferative disease (especially CLL) in ITGN.^{336,337} A few cases of ITGN have been associated with HIV (with or without concomitant hepatitis C). Also of note is the absence of DNAJB9 protein in ITG disease.

Given the high association with hematologic disorders, chemotherapy directed against the underlying lymphoproliferative disorder, if identified, is paramount. In such cases, even a partial remission of the hematologic disease has led to complete renal remission. The few cases of reported HIV-positive ITGN patients have shown variable responses to antiretroviral therapy, corticosteroids, and RAAS inhibitions.³³⁸ Data on allograft fate after kidney transplantation are sparse given the rarity of disease. Recurrence has ranged from 25% to 50% in small case series.³³⁹

NAIL-PATELLA SYNDROME (HEREDITARY OSTEO-ONYCHODYSPLASIA)

Nail-patella syndrome (NPS), inherited in an autosomal dominant manner, affects tissues of ectodermal and mesodermal origin and presents with nail, skeletal, ocular, and renal anomalies. The condition, described in 1948, is characterized by symmetric nail dysplasia and patellar aplasia or hypoplasia. The presence of triangular nail lunulae is pathognomonic. Thumb nails are maximally affected; the nails may be absent, hypoplastic, or dystrophic with ridges and/or pits. Other skeletal abnormalities include dysplasia of the elbow joints, posterior iliac horns, and foot deformities. Ocular anomalies include microcornea, sclerocornea, congenital cataracts, iris processes, pigmentation of inner margin of the iris, and open-angle glaucoma.³⁴⁰ Involvement of the kidneys is variable, in up to 40% patients. Manifestations appear in children and young adults and include proteinuria, hematuria, hypertension, or edema. Steroid-resistant nephrotic syndrome and progressive kidney failure are a late feature.^{341,342} Congenital malformations of the urinary tract and nephrolithiasis are also more common in these patients. Cases with kidney lesions typical of NPS, but without skeletal abnormalities, have been reported.³⁴³

Findings on LM include focal segmental glomerulosclerosis, segmental thickening of the glomerular capillary wall, and mild mesangial hypercellularity.³⁴⁴ Ultrastructural studies show a thickened GBM that contains irregular lucencies, imparting a “moth-eaten” appearance (Fig. 30.18A). Intramembranous fibrils are seen, with the 65 nm periodicity of collagen revealed by phosphotungstic acid stains, corresponding to the distribution of intramembranous lucencies (see Fig. 30.18B). These must be distinguished from the occasional collagen fibrils that might accumulate nonspecifically in the mesangium in conditions associated with glomerular sclerosis.³⁴⁴

The genetic locus for NPS is on chromosome 9q34 and results from mutations in the LIM homeodomain protein *LMX1B* gene that is transmitted in an autosomal dominant

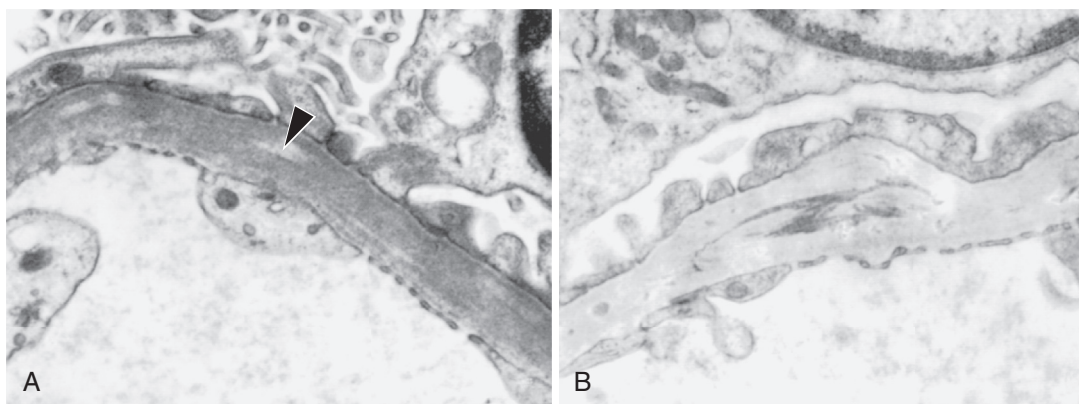


Fig. 30.18 Nail-patella syndrome. (A) Routine electron micrograph showing thickening of a glomerular basement membrane with focal irregular internal lucencies (arrowhead) (×15,000). (B) Phosphotungstic acid-stained electron micrograph demonstrating the characteristic banded collagen fibrils within the rarefied segments of glomerular basement membrane (×15,000).

pattern and encodes a transcription factor. LMX1B plays a central role in dorsoventral patterning of the vertebrate limb.^{345,346} In mice, genetic ablation of *LMX1B* leads to loss of Nphs2 expression, as well as loss of expression of the GBM collagens Col4a3 and Col4a4, suggesting that Lmx1b potentially acts as a common upstream regulator of these genes.³⁴⁷ There is no specific treatment; patients with renal failure have been successfully transplanted.³⁴⁸

LIPOPROTEIN GLOMERULOPATHY

Lipoprotein glomerulopathy is a rare kidney disease characterized by proteinuria, progressive kidney failure, abnormal plasma lipoprotein profile indicating type III hyperlipoproteinemia, markedly elevated serum apolipoprotein E (ApoE), and lamellated lipoprotein thrombi in glomerular capillaries without foam cells.³⁴⁹ Characteristic abnormalities, in common with type III hyperlipoproteinemia, include high levels of both cholesterol and triglycerides, accumulation of incompletely catabolized triglyceride-rich lipoproteins, palmar xanthomas, and rapidly progressive atherosclerosis.^{350,351}

ApoE is a 34kDa apolipoprotein that binds to a LDL receptor to facilitate the catabolism of triglyceride-rich lipoproteins.³⁵² Absence or dysfunction of ApoE impairs clearance of chylomicrons, VLDL, and LDL. Patients with lipoprotein glomerulopathy have heterozygous variants in *APOE* (location 19q13.2) that alter the protein's α -helix to reduce its helicity and structural stability.³⁵³ Several of the 18 reported mutations in *APOE* transcribe into the protein domains binding to the LDL receptor and/or heparan sulfate proteoglycan.³⁵⁴ While four major variants share substitutions of proline (*APOE* Sendai, *APOE* Chicago, *APOE* Guangzhou, and *APOE* Osaka/Kurashiki), nonproline substitutions (*APOE* Kyoto, *APOE* Tsukuba, *APOE* Las Vegas) have also been reported.^{353,355}

Most patients with lipoprotein glomerulopathy are asymptomatic and diagnosed following evaluation of cause for proteinuria detected on routine screening. While the severity of proteinuria is variable, most patients develop nephrotic syndrome or nephrotic-range proteinuria. The median age at presentation was 35 years (range 7–72 years); 17% of cases were detected in childhood or adolescence, and 60% in young adulthood (18–45 years).³⁵⁶ The majority of patients developed nephrotic-range proteinuria or nephrotic syndrome.^{356,357} One-half of patients eventually develop end-stage kidney disease (ESKD) over 1 to 27 years.³⁵⁶ Extrarenal comorbidities and disease association are rarely reported including thrombosis, splenomegaly, aortic aneurysm, pleural effusion, thalassemia, psoriasis, and atypical hemolytic uremic syndrome.³⁵⁶

A diagnosis of lipoprotein glomerulopathy is made in patients with proteinuria; type III pattern of hyperlipoproteinemia (predominance of triglycerides; high levels of VLDL and intermediate-density lipoprotein, IDL; elevated VLDL to triglyceride ratio) on ultracentrifugation and agarose gel electrophoresis; increased serum ApoE levels (usually >10 mg/dL); normal lecithin-cholesterol acyltransferase (LCAT) activity; and characteristic findings

on kidney biopsy.³⁵⁰ The diagnosis is most likely in Japanese or Chinese adults. Where possible, the diagnosis should be confirmed by sequencing of *APOE*, which also helps distinguish it from familial dysbetalipoproteinemia (type III hyperlipoproteinemia), characterized by homozygosity for ApoE E2 isoform (E2/E2 genotype).³⁵¹

Light microscopy of kidney biopsy reveals marked dilatation of glomerular capillary lumina by pale eosinophilic granular and vacuolar thrombi.³⁵⁸ Positive staining for oil red O or Sudan red and negative stain for silver or periodic-acid Schiff confirms the presence of lipids within the thrombi. Foam cells, which are characteristic of lipidosis, are rarely seen. Immune-electron microscopy, if performed, shows that these thrombi comprise lipoproteins surrounded by ApoE.³⁵⁹ On electron microscopy, the glomerular capillary thrombi appear to be composed of granules and vacuoles of various sizes, which form concentric lamellate, like a fingerprint or sandstone structure.^{350,360}

There are no established therapies for the condition. Therapy with immunosuppressive agents is not useful. Experts recommend intensive lipid-lowering therapy along with conventional antiproteinuric strategies while monitoring serum levels of lipids, electrolytes, and eGFR and targeting reduction in proteinuria. Since the disease recurred in five of six allografts whose outcomes were reported,³⁵⁶ kidney transplantation was not considered a satisfactory option. However, outcomes may be more promising with newer management strategies.

LIPODYSTROPHY

Lipodystrophy syndromes comprise a group of conditions characterized by absence of body fat; low levels of an adipocyte-derived hormone or cytokine leptin; severe hypertriglyceridemia, with or without insulin resistance; diabetes; and hepatic steatosis.³⁶¹ On the basis of underlying genetic basis and distribution of fat loss, four major categories are recognized: congenital generalized lipodystrophy, acquired generalized lipodystrophy, familial partial lipodystrophy, and acquired partial lipodystrophy.^{361–363} Each category has subtypes based on etiology and associated findings. Additional categories include progeroid disorders, autoimmune disorders, lipodystrophy in patients with HIV, and localized lipodystrophy following use of injectable drugs.^{361–363}

Kidney disease is most prevalent in patients with generalized lipodystrophy but is also described in certain partial forms.^{361,364,365} Proteinuria is the most common finding, which might be associated with hematuria, nephrotic syndrome, FSGS, and/or progressive CKD. Certain partial forms of lipodystrophy are associated with membranoproliferative glomerulonephritis (MPGN) secondary to a C3-nephritic factor.³⁶⁶

While the mechanisms of proteinuria remain unclear, it is hypothesized that the metabolic abnormalities in lipodystrophy alter renal hemodynamics, leading to glomerular hyperfiltration, glomerulomegaly, and adaptive FSGS.³⁶⁷ Animal experiments show that lipid accumulation is associated with increased expression of

sterol regulatory element binding protein 1 (SREBP-1), which induces glomerulosclerosis and proteinuria.^{368,369} Patients with acquired partial lipodystrophy (APL), also termed Barraquer-Simons syndrome, may have a polyclonal immunoglobulin C3-nephritic factor that causes persistent amplification of the alternative complement pathway and low complement C3 levels. The deposition of C3 into the kidney glomeruli leads to a membranoproliferative pattern of glomerulonephritis, as either dense deposit disease or C3 glomerulonephritis.³⁶⁶

Lipodystrophy should be suspected in lean and muscular subjects presenting with diabetes mellitus, hypertriglyceridemia, nonalcoholic fatty liver disease, polycystic ovarian syndrome, or amenorrhea.^{361,370} The diagnosis is clinical, based on the finding of loss of body fat or its altered distribution. Body composition analysis by skinfold measurements, dual-energy x-ray absorptiometry and whole-body magnetic resonance imaging, and demonstration of low levels of leptin (<12.0 ng/mL) confirm the diagnosis. Patients with early onset disease and positive family history should undergo genetic testing.³⁷⁰ The most common familial partial lipodystrophy, namely Dunnigan lipodystrophy (FPLD2), is characterized by fat loss (lipoatrophy) from the limbs, abdomen, and thorax; excessive subcutaneous fat on the face, chin and supraclavicular areas; and insulin resistance.³⁷¹ The diagnosis is suspected in nonobese adolescent girls or women with severe hyperandrogenism, acanthosis nigricans, and symptoms of polycystic ovary syndrome (type A insulin resistance), associated with altered fat distribution, similar family history, hypertriglyceridemia, and low serum HDL cholesterol.³⁷¹

GLOMERULAR DISEASES ASSOCIATED WITH DRUGS

Drug-induced glomerular diseases can be classified mechanistically into two categories: direct cellular toxicity to podocytes, endothelium, or mesangial cells and immune-mediated injury.³⁷² The importance of their recognition cannot be overemphasized; cessation of therapy with the offending agent is usually associated with reversal of the lesions and clinical and biochemical improvement. [Table 30.8](#) lists drugs that cause glomerular injury by these mechanisms and few examples of these distinct categories are discussed as follows.

Drug-induced podocytopathies present with either nephrotic syndrome or nephrotic-range proteinuria, with or without AKI.³⁷³ Kidney biopsy might show MCD, focal segmental glomerulosclerosis (FSGS), or collapsing glomerulopathy. Medications that are implicated include nonsteroidal antiinflammatory drugs (e.g., indomethacin and celecoxib), bisphosphonates (e.g., pamidronate), sirolimus, androgenic anabolic steroids, interferon, lithium, and tyrosine kinase inhibitors.

Mesangial injury leading to IgA nephropathy has been described following therapy with vancomycin,

ceftriaxone, cyclosporine, carbamazepine, acetaminophen, amiodarone, furosemide, and immune-checkpoint inhibitors (ipilimumab, pembrolizumab, nivolumab). While heavy cigarette smoking is associated with vascular injury and hypertensive arteriosclerosis, it typically induces nodular glomerulosclerosis. Cholesterol emboli in glomeruli are reported following therapy with warfarin, streptokinase, or alteplase.

Drugs are among the leading causes of thrombotic microangiopathy, particularly in adults. Clinical suspicion, based on microangiopathic hemolytic anemia, thrombocytopenia, and AKI, is confirmed on kidney biopsy by demonstration of endothelial swelling, glomerular and/or arteriolar thrombosis, and mesangiolysis. Medications implicated include thienopyridines (clopidogrel, ticlopidine); calcineurin and mTOR inhibitors (tacrolimus, cyclosporine, sirolimus, everolimus); quinine; chemotherapeutic agents, including antiangiogenic monoclonal antibodies (e.g., bevacizumab); gemcitabine; vincristine; mitomycin C; palbociclib; proteasome inhibitors; interferon; and opioids. Postulated mechanisms include direct endothelial cell injury, inhibition of the VEGF, and induction of anti-ADAMS13 or antiplatelet antibodies. Despite normal serum complement and absence of genetic defects in complement regulatory proteins, some patients respond to therapy with eculizumab, indicating a role of complement activation.

The most commonly encountered form of drug-induced autoimmunity has been MN, encountered with penicillamine, gold salts (used for managing rheumatoid arthritis), captopril, lithium, and NSAIDs (celecoxib), and more recently with adalimumab (used in rheumatic diseases), chemotherapeutic agents like epidermal growth factor receptor tyrosine kinase inhibitors (gefitinib) and immune-checkpoint inhibitors (nivolumab), and organic mercurials (in skin-lightening creams).³⁷⁴ Immune-mediated injury may also follow drug-induced autoimmunity, which is triggered in <1% of cases exposed to any medication or its metabolite. Such reactions are usually idiosyncratic (i.e., are unpredictable and not related to the dose or mechanism of action of the drug).³⁷³ Symptomatic kidney pathology, seen in <5% of such cases, might be accompanied by skin and systemic manifestations of autoimmunity. While therapy with sulfadiazine, procainamide, hydralazine, and quinidine is associated with lupus-like lesions, ANCA-associated vasculitis (AAV) is reported after exposure to levamisole-adulterated cocaine, antithyroid drugs, hydralazine, and immune-checkpoint inhibitors.

In summary, drug-induced glomerular disease encompasses a wide range of pathophysiologic mechanisms, histologic presentations, and clinical courses. Notably, a single drug may give rise to distinct histologic patterns of injury, underscoring the complexity of its manifestations.

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Table 30.8 Examples of Drug-Induced Glomerular Injury

Lesion	Histology or Mechanism	Examples
Podocytopathy with proteinuria, or glomerulonephritis	Minimal change disease; focal segmental glomerulosclerosis, collapsing glomerulopathy	Nonsteroidal antiinflammatory drugs: indomethacin, celecoxib Lithium Bisphosphonates: pamidronate, zoledronate mTOR inhibitors: everolimus, sirolimus Interferon Androgenic anabolic steroids ALK inhibitors: crizotinib, lorlatinib, alectinib Immune-checkpoint inhibitors targeting CTLA4: Ipilimumab, nivolumab PD-1 inhibitors: pembrolizumab, cemiplimab, nivolumab Tyrosine kinase inhibitors: sorafenib, sunitinib, axitinib, pazopanib, lenvatinib VEGF-A monoclonal antibody: bevacizumab VEGFR2 monoclonal antibody: ramucirumab VEGF-Trap (fusion protein): aflibercept
	IgA nephropathy	Vancomycin, carbamazepine, ceftriaxone, metronidazole, cyclosporine, acetaminophen, amiodarone, furosemide, nivolumab, ipilimumab, pembrolizumab
	Miscellaneous lesions	Allopurinol, atorvastatin, cocaine/levamisole, denosumab, hydralazine, isotretinoin, phenytoin, febuxostat, foscarnet, methamphetamine, sofosbuvir in kidney transplant recipients, protease inhibitors
	Thrombotic microangiopathy	Antitumor antibiotics: mitomycin C Antimetabolites: gemcitabine, pentostatin Inhibitors of cyclin-dependent kinases CDK4 and CDK6: palbociclib BCR-ABL tyrosine kinase inhibitors: imatinib, dasatinib Protease inhibitors: bortezomib, carfilzomib Tyrosine kinase inhibitors: sorafenib, sunitinib, axitinib, pazopanib, lenvatinib Other chemotherapeutic agents: vincristine, proteasome inhibitors, bleomycin, cisplatin, oxaliplatin, and carboplatin Calcineurin inhibitors, sirolimus, everolimus Others: interferon, opioids, valproic acid, IV immunoglobulin G VEGF-A monoclonal antibody: bevacizumab
Autoimmune	Vascular smooth muscle cell dysfunction (inhibition of angiogenesis or VEGF deficiency)	
	Antibody mediated	Quinine, interferon, thienopyridines, oxaliplatin, quetiapine, gemcitabine, muromonab-CD3, penicillin, sulfisoxazole, trielina
	Vasculitis; often ANCA-associated (pauci-immune)	Levamisole (with or without cocaine) Antithyroid drugs: propylthiouracil, carbimazole and methimazole Hydralazine Antimicrobials: minocycline, antituberculosis drugs Disease-modifying agents for rheumatic diseases: penicillamine, sulfasalazine, allopurinol Psychoactive agents: cocaine Immune-checkpoint inhibitors: ipilimumab, pembrolizumab, nivolumab, tremelimumab
	Lupus-like lesions	TNF- α inhibitors Procainamide Hydralazine Quinidine Low risk: TNF- α inhibitors, carbamazepine, PTU, methyl dopa, captopril, acebutolol, chlorpromazine, isoniazid, minocycline, ipilimumab, nivolumab
	Membranous nephropathy	Gold therapy, penicillamine, and bucillamine Lithium Captopril NSAIDs, celecoxib Gefitinib, nivolumab, adalimumab Mercaptopropionyl, chlopropamide, aprotinin

ALK, Anaplastic lymphoma kinase; BCR-ABL, breakpoint cluster region-tyrosine protein kinase ABL-1 gene; CDK, cyclin-dependent kinase 4 and 6; CTLA-4, cytotoxic T-lymphocyte-associated protein 4; PD-1, programmed cell death protein 1; VEGF, vascular endothelial growth factor; VEGFR, VEGF receptor.

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